



Chapter 7 : Lithography Technology- I

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Contents

- **Basic concepts**
- **Lithography technologies/Radiation/matter interaction**
- **Lithographic materials**
- **Key Factors in Optical Lithography**





Lithography



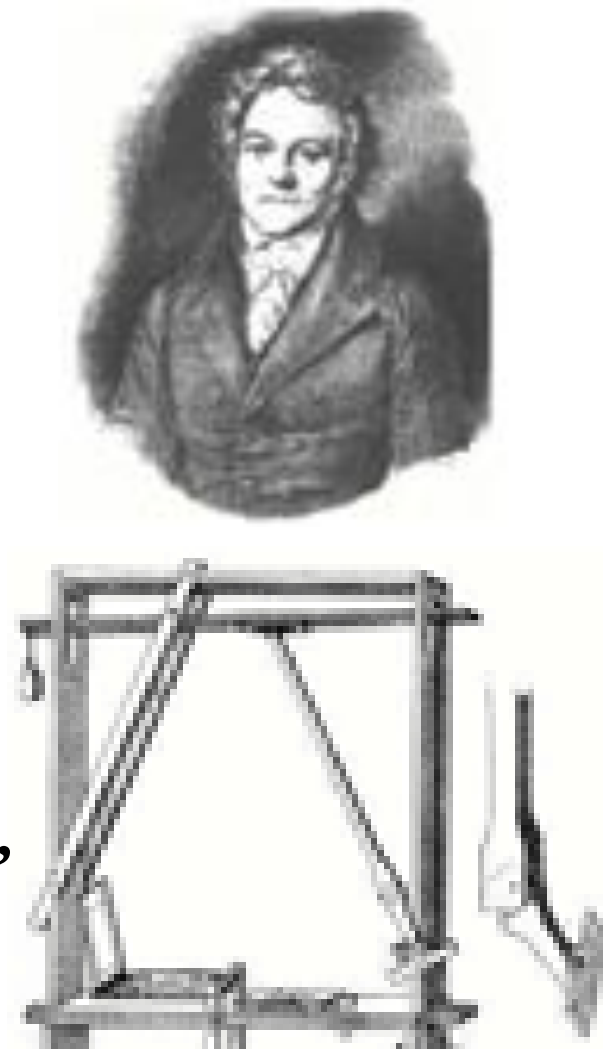
Marc Chagall
Creation, 1887

Origins of Lithography

Invented by Alois Senefelder, 1796
“chemical printing”

- Limestone marked with oil-based substance
- Acid burns image into surface
- Gum-arabic solution adheres only to non-oiled area
- During printing, water adheres to gum-arabic areas, while oily ink (soap, wax, oil, lampblack) is transferred to paper

lithos = stone
graphy = writing



Ultimate Research Lithography

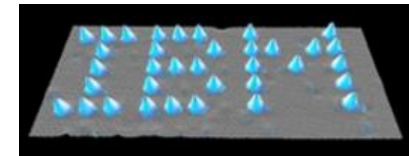
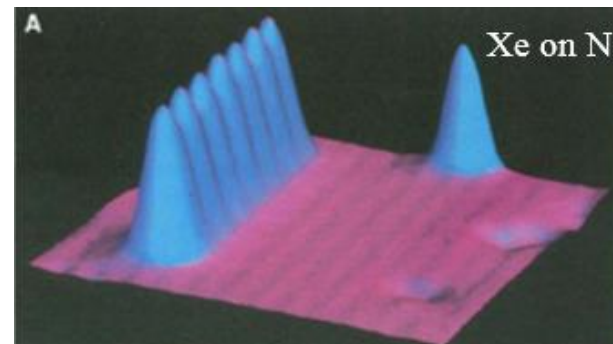
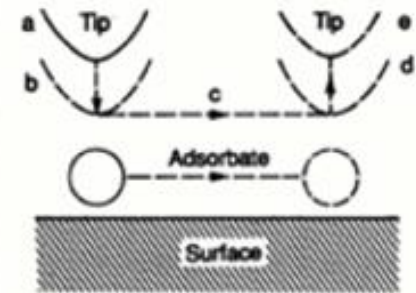
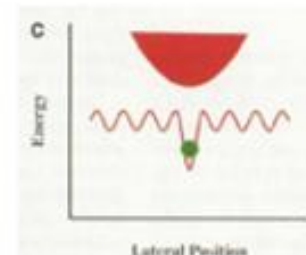
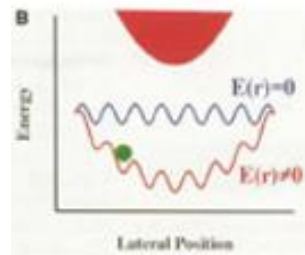
- Pick and place atoms
- STM (atomically sharp)
- UHV
- cryogenic temperatures
- Atomically clean surface

- Resolution:

- ~ 0.3 nm (1 atom)

- Throughput:

- 1 atom/min
 - ≈ 0.017 pixels/sec

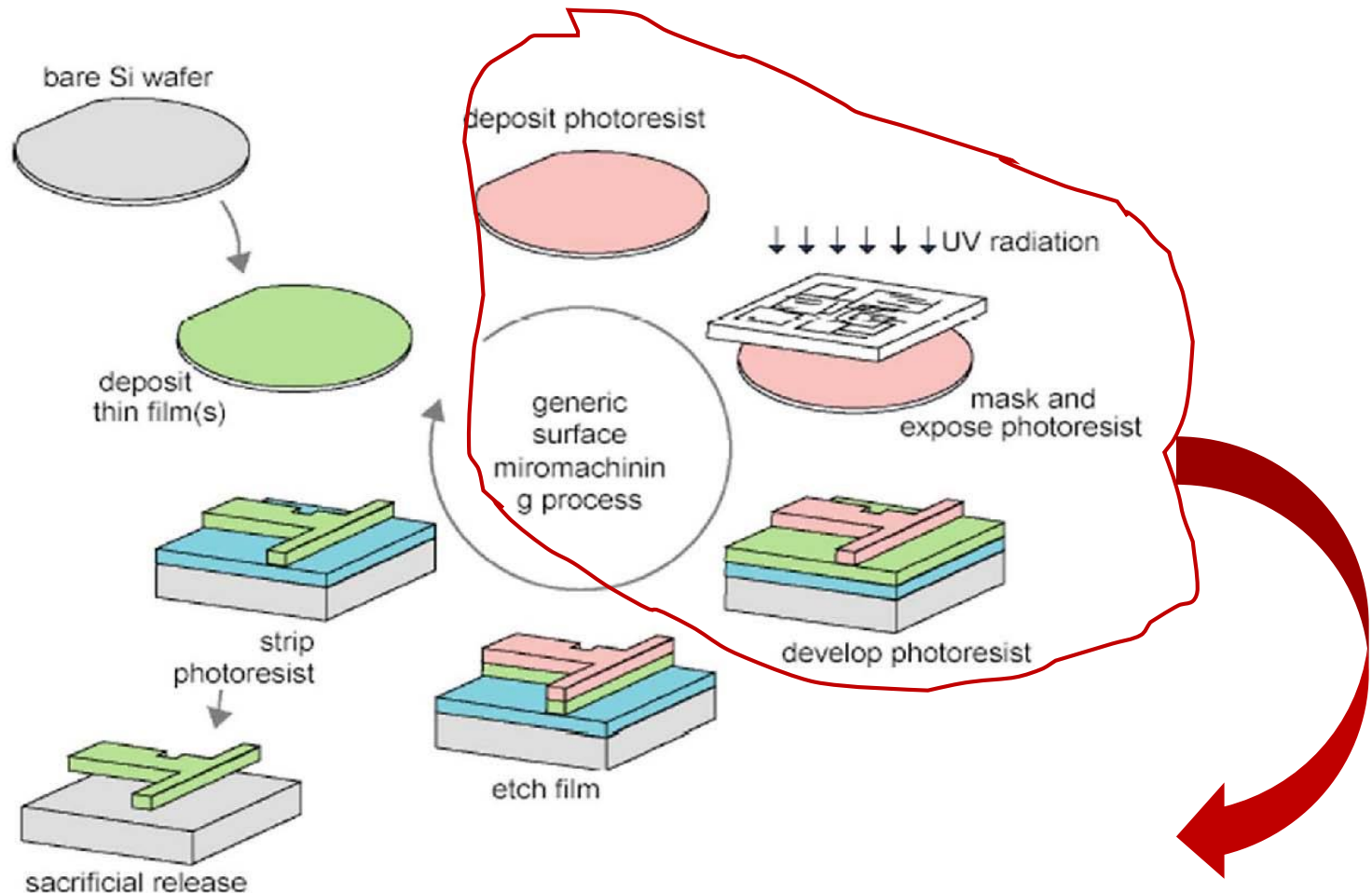


D. Eigler, IBM

➡ Not General process



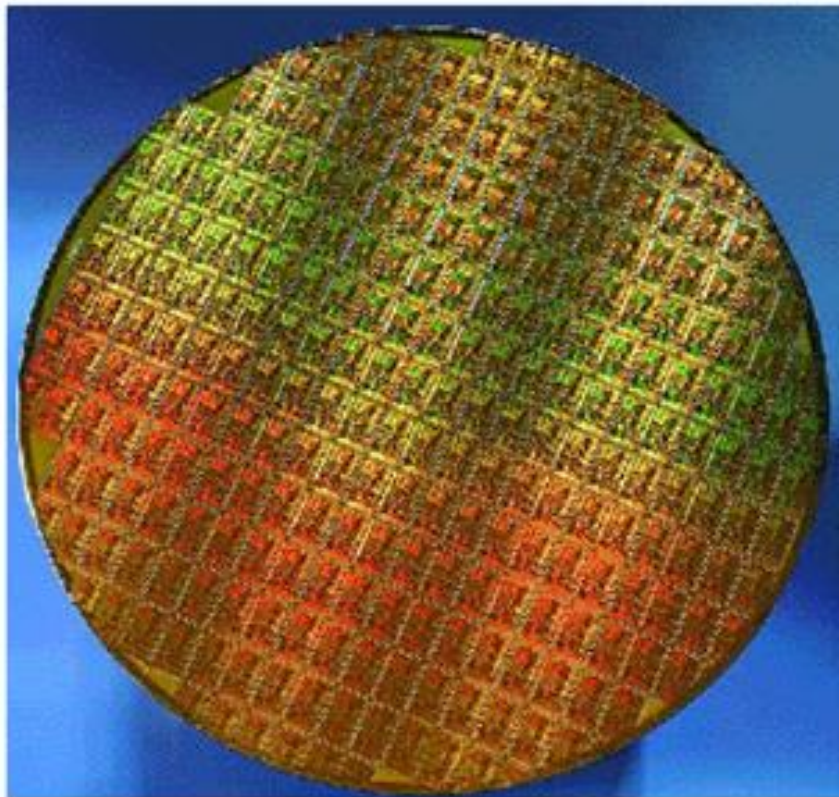
Key process in Planar Fabrication Process (Generic)



Cost > 60%!!



“Chips”



Current Production Lithography

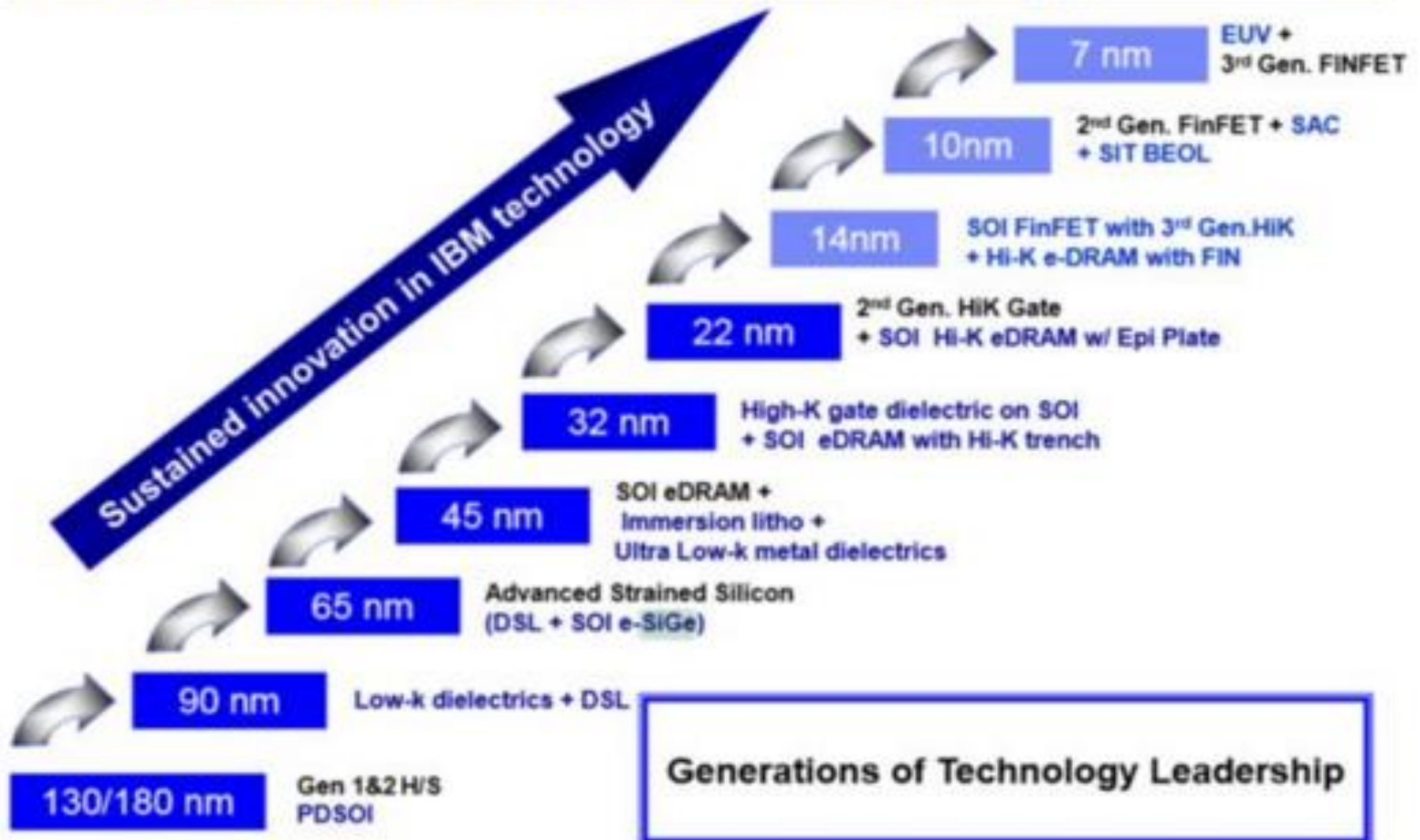
- 5 nm EUV Lithography(TSMC)
- Chemically amplified resist
- 193 nm step & scan exposure
- Si wafer substrate
 - previously patterned (usually)
- Resolution:
 - $\sim \lambda/4$
 - ~ 50 nm (< 30 nm using **tricks**)
- Throughput:
 - 100 wafers/hour (300 mm)
 - $\approx 10^{11}$ pixels/sec.





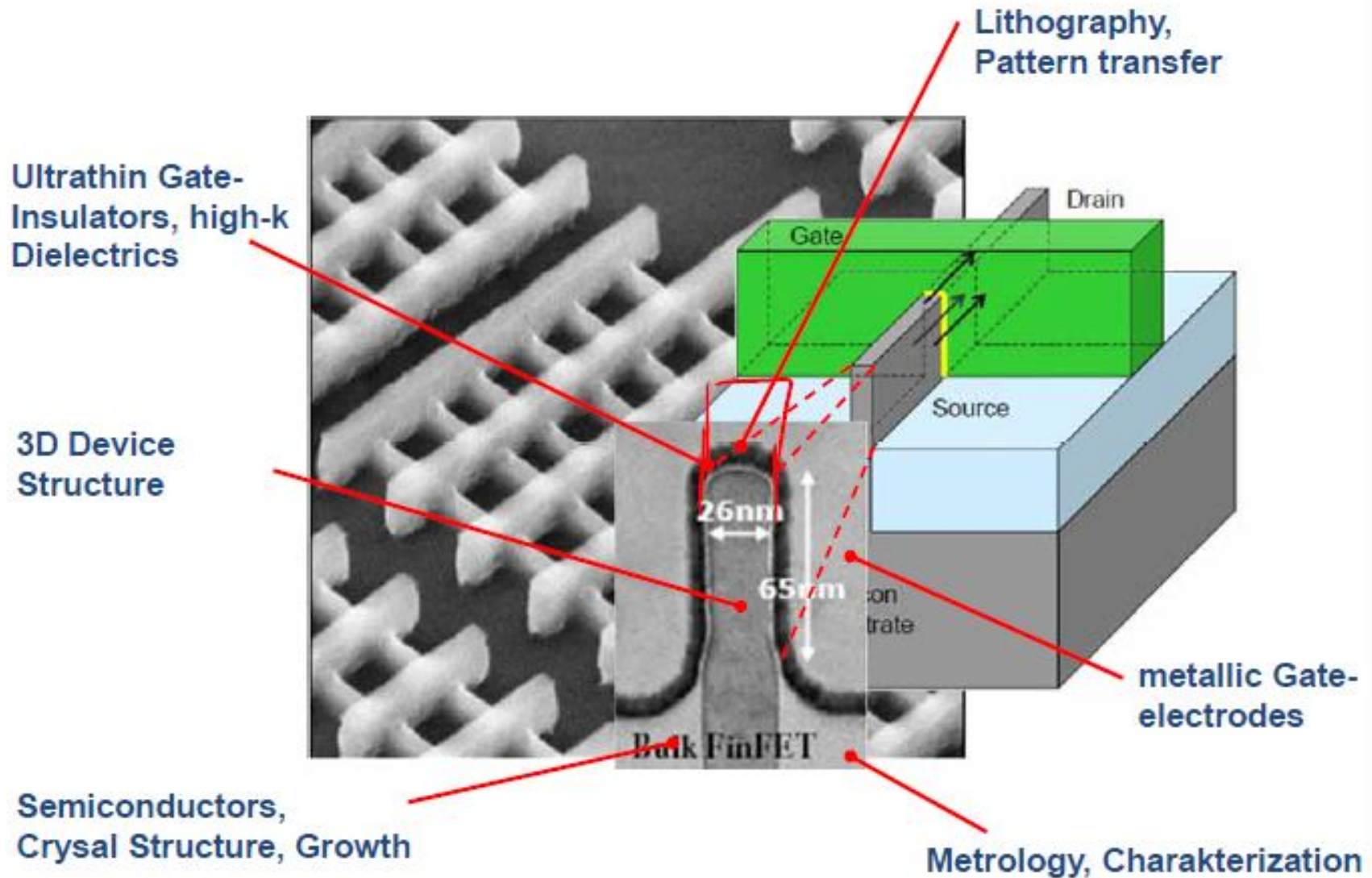
Ten Generations of IBM SOI Technology

IBM





Current Production Lithography



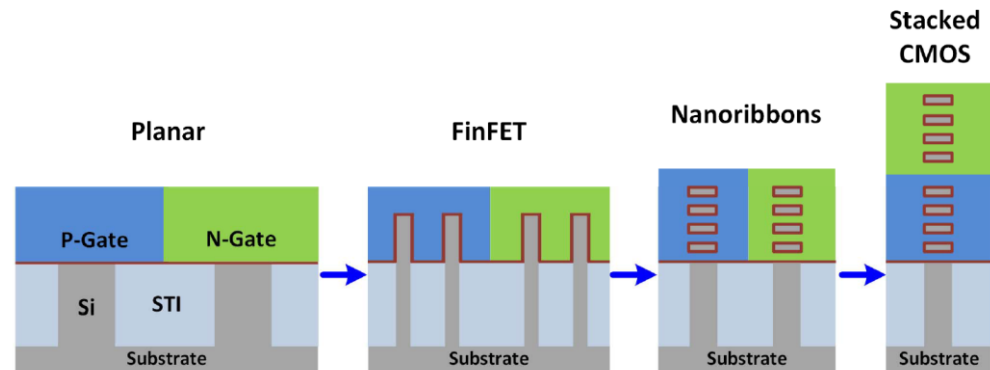
Current Production Lithography

- State of the art lithography tool (production)2020:

5nm EUV –

~ 5 nm minimum linewidth

Example: 麒麟990 5G
7nm EUV



- **Lithographic process = communication**

Exp: 193nm ArF Immersion lithography

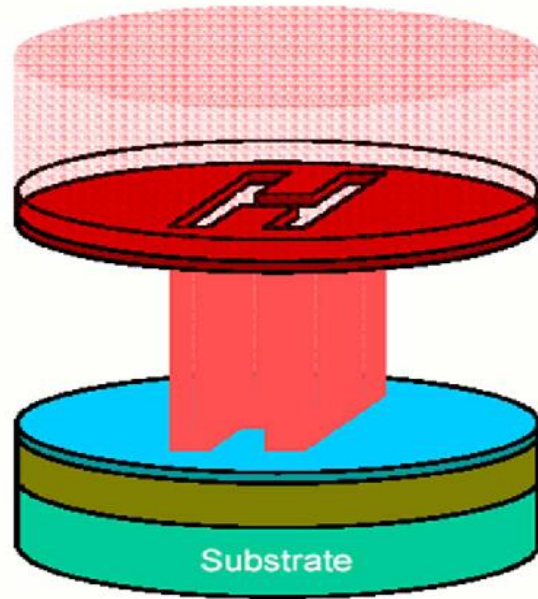
Contents

- Basic concepts
- Lithography technologies and Radiation/matter interaction
- Lithographic materials
- Key Factors in Optical Lithography





Generic Lithographic Process

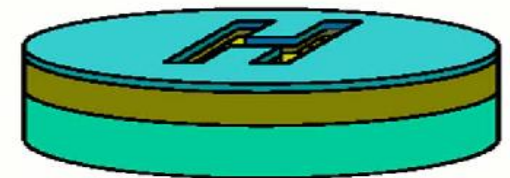
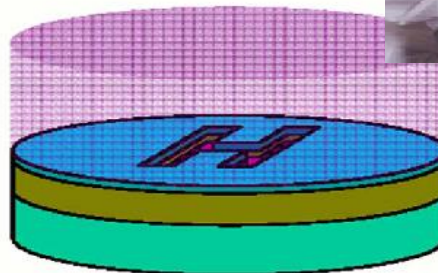
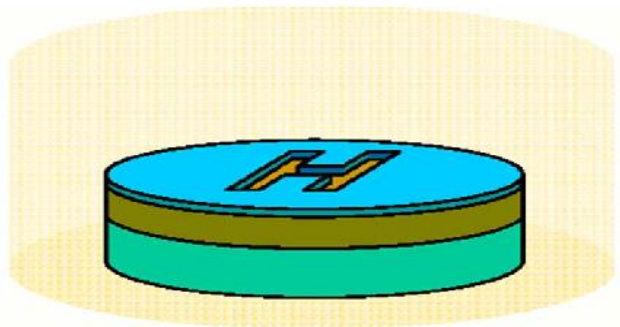


Radiation (Exposure)

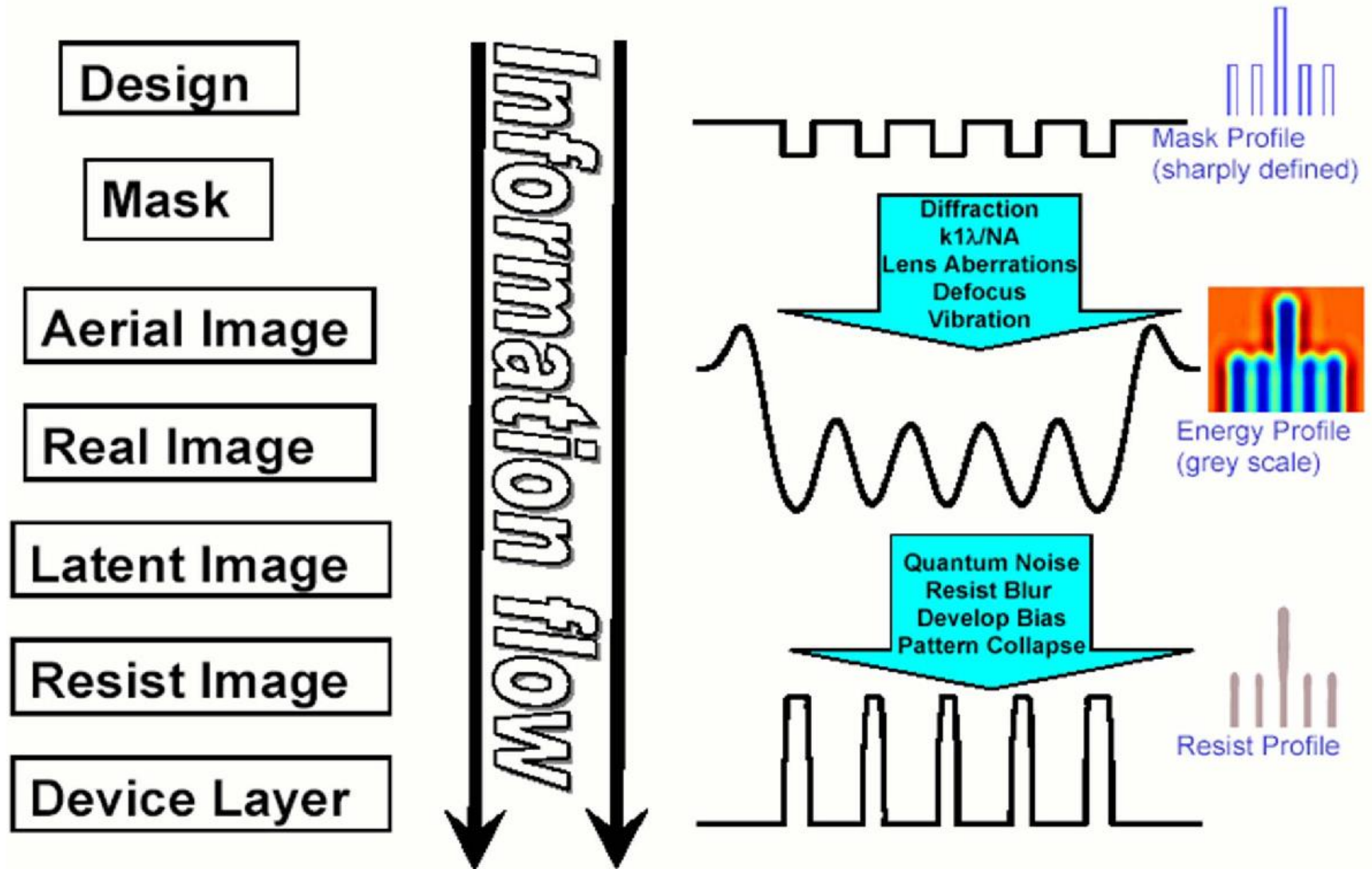
Mask (Alignment)

Photoresist (spin-Coating)

Thin Film (Deposition)

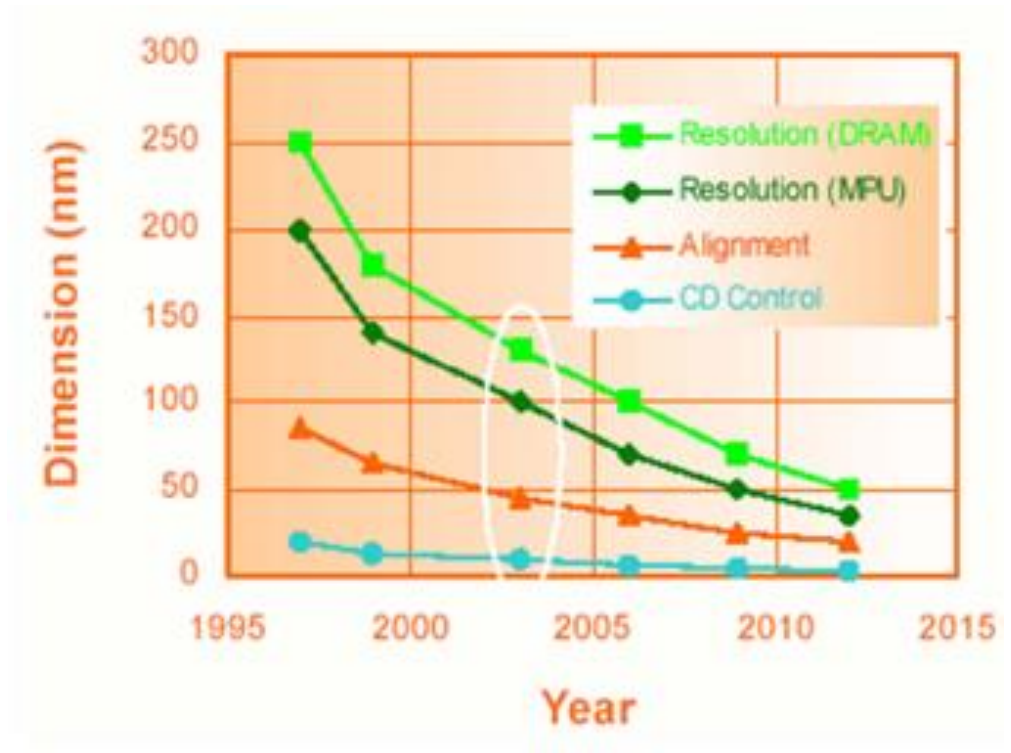


Lithographic Process as Information Flow



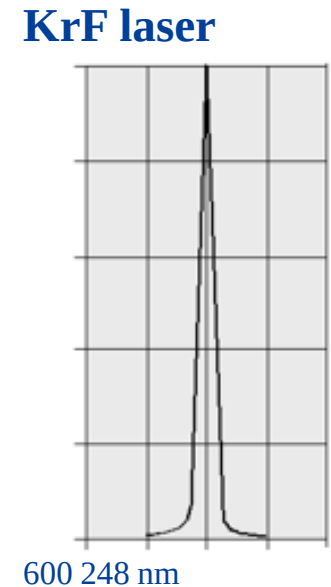
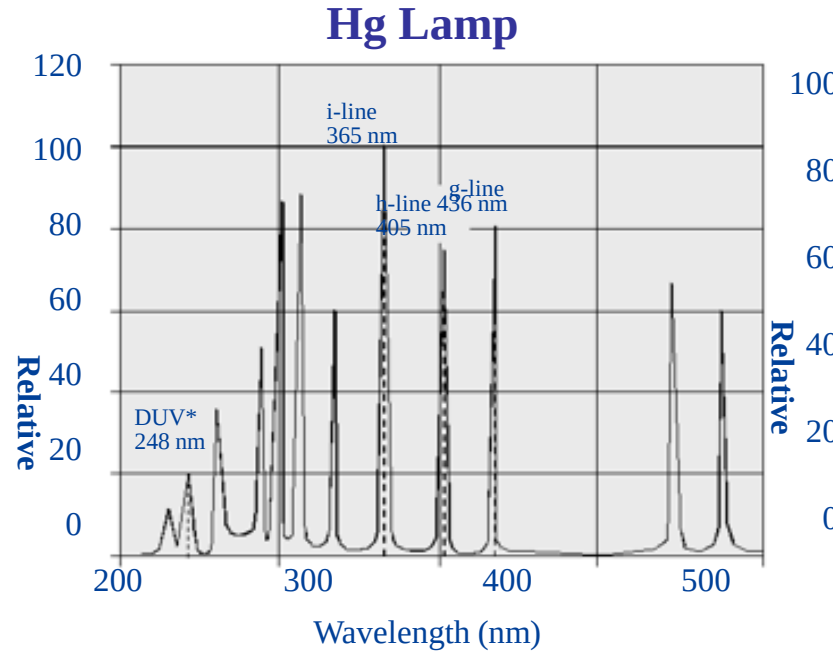
Key Lithography Metrics

- Resolution
 - Minimum feature size
 - Dimensional control
 - across sample
 - sample to sample
- Contrast
- Throughput
- Placement accuracy
 - overlay
 - stitching



光刻要素之一：Sources

- Hg arc lamp
 - G-line: 436 nm
 - H-line: 405 nm
 - I-line: 365 nm
- Excimer laser
 - KrF: 248 nm
 - ArF: 193 nm
- Ultra-excimer
 - F₂: 157 nm



- Filters used to limit exposure wavelength
- Intensity uniformity must be better than several % over collected area
- Source aging dictates periodic calibration (spectral exposure meter)



g-line

- 波长：436nm
- 制程：250-800nm

i-line

- 波长：365nm
- 制程：250-800nm

KrF

- 波长：248nm
- 制程：130-180nm

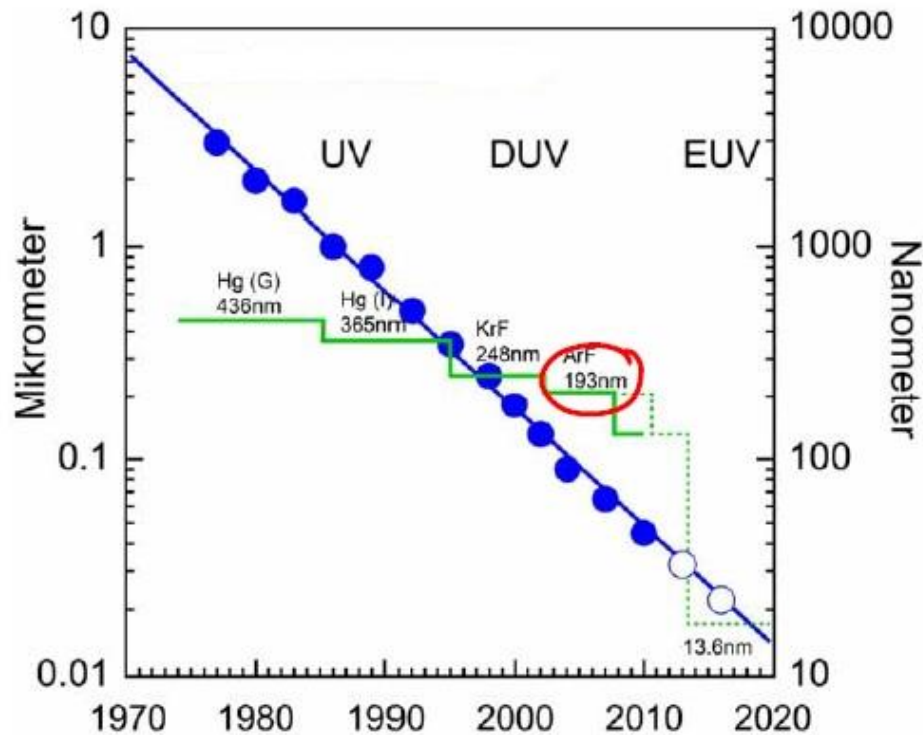
ArF

- 波长：193nm
- 制程：65-130nm（步进投影式）/22-45nm（浸没步进式）

EUV

- 波长：13.5nm
- 制程：7-22nm

Optical Lithography



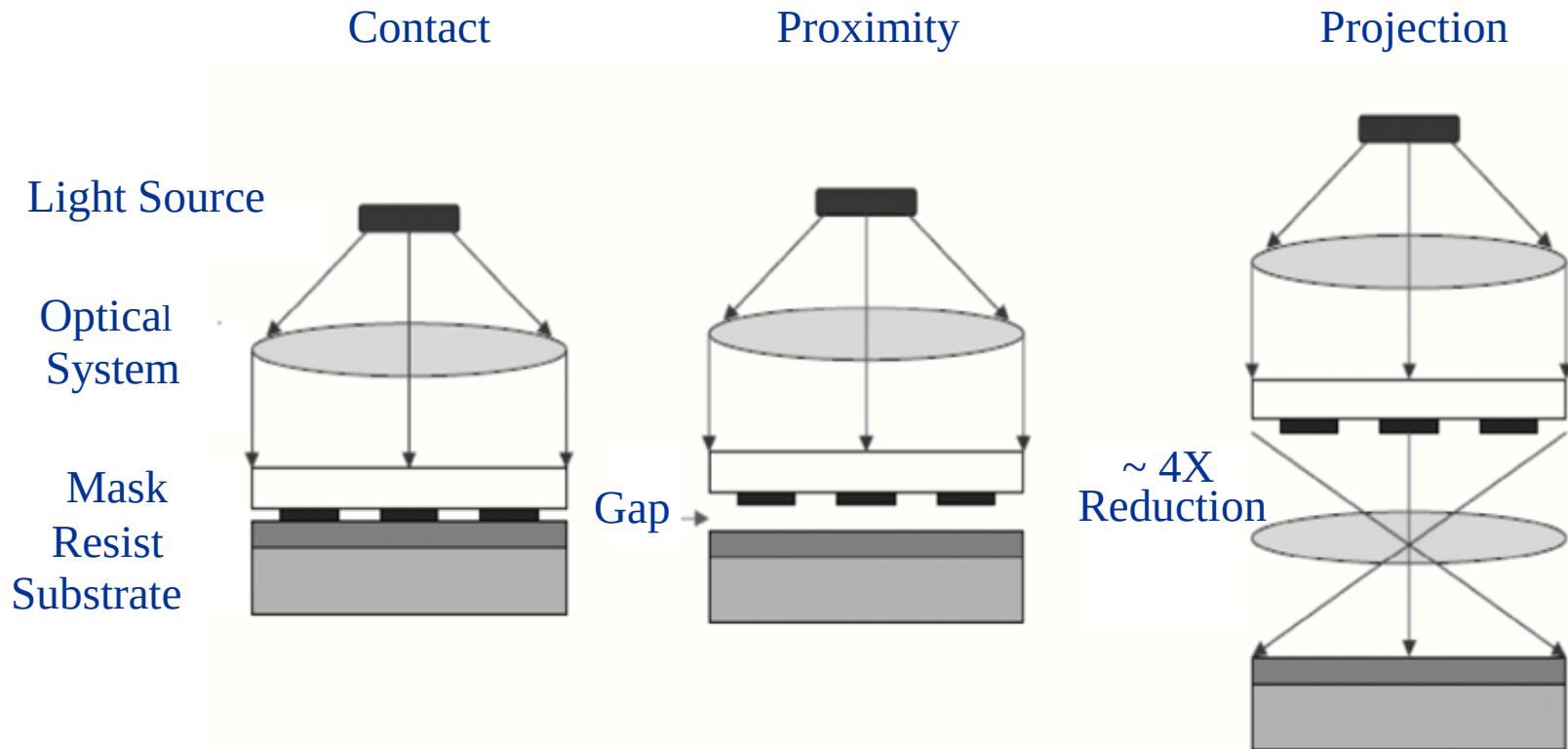
Since several years: critical dimensions significantly smaller than wavelength of light source

Use resolution enhancement techniques(tricks)

- Off-axis illumination
- Phase shift mask
- Optical proximity correction



要素之二：Exposure Systems

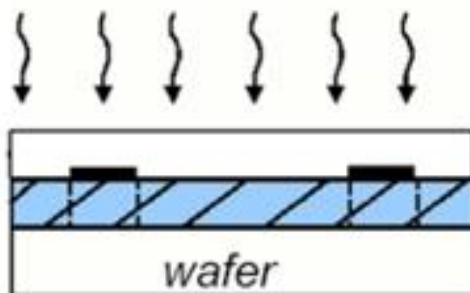




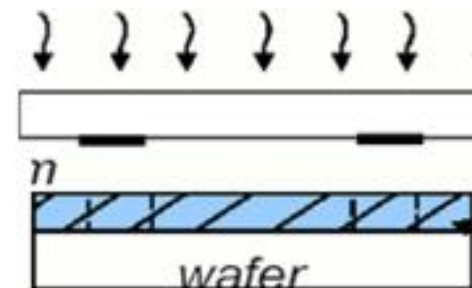
Proximity Printing with a Mask Aligner



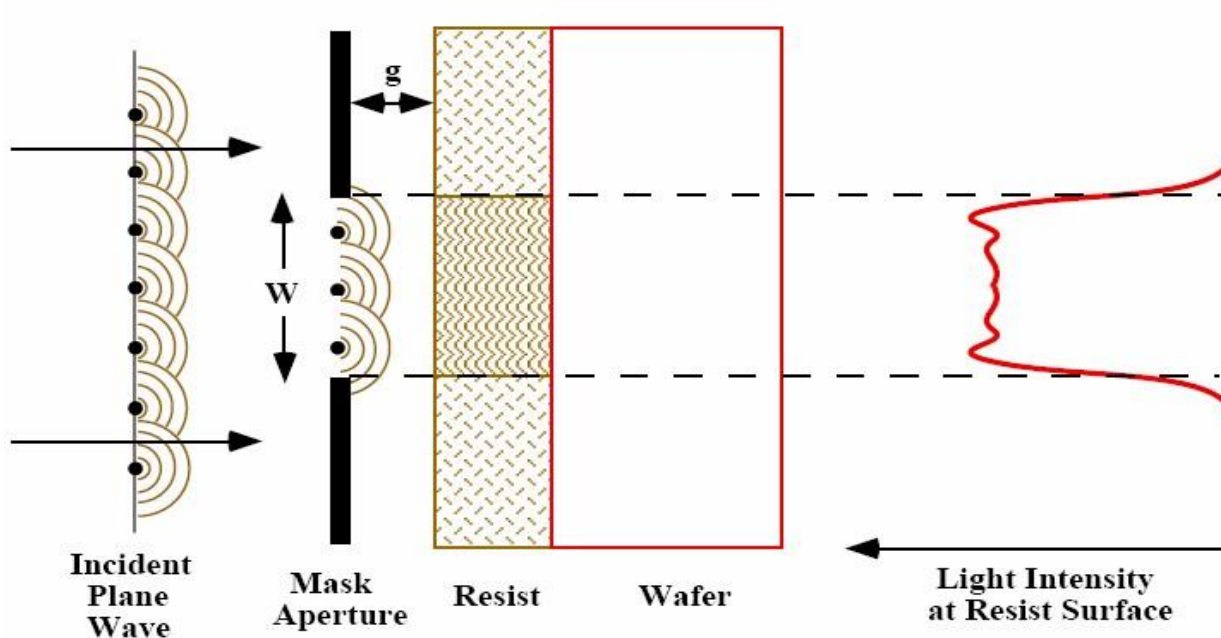
In theory



In practice



Contact and Proximity



Fresnel衍射理论适用的
的间隔范围:

$$\lambda < g < \frac{W^2}{\lambda}$$

最小分辨尺寸

$$W_{\min} \approx \sqrt{\lambda g}$$

$$\lambda = 400\text{nm}$$

$$g = 50 \mu\text{m}, W_{\min} \cong 4.5 \mu\text{m}$$

$$g = 2 \mu\text{m}, W_{\min} \cong 0.9 \mu\text{m}$$

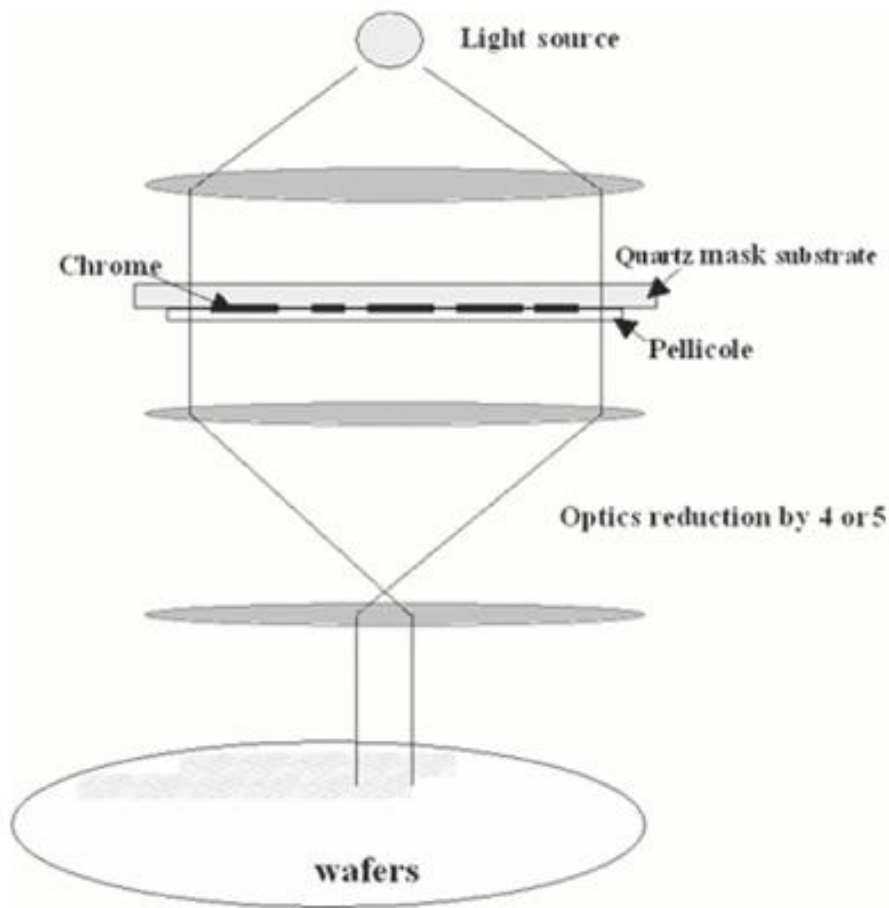
$$\lambda = 250\text{nm}$$

$$g = 50 \mu\text{m}, W_{\min} \cong 3.5 \mu\text{m}$$

$$g = 2 \mu\text{m}, W_{\min} \cong 0.7 \mu\text{m}$$



Projection Printing



Demagnification: nX

• **10X**

• **5X**

• **4X**

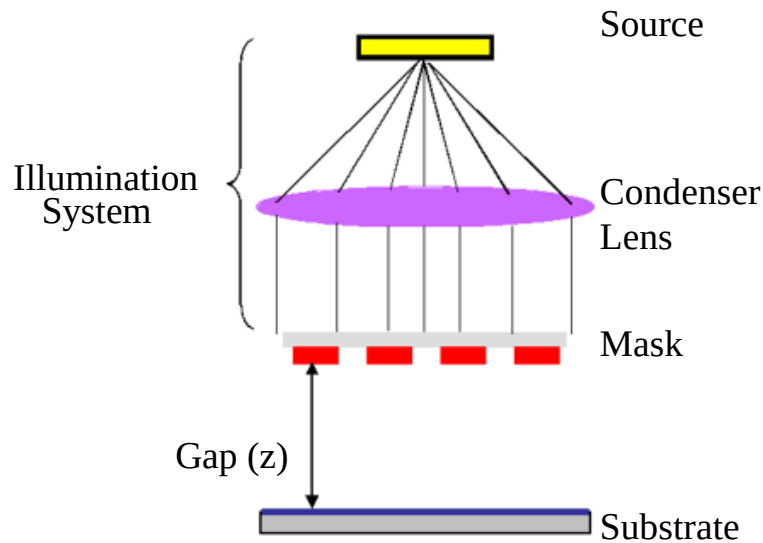
• **Very high resolution (deep UV photons)**

• **Complicated optics**

• **Expensive**

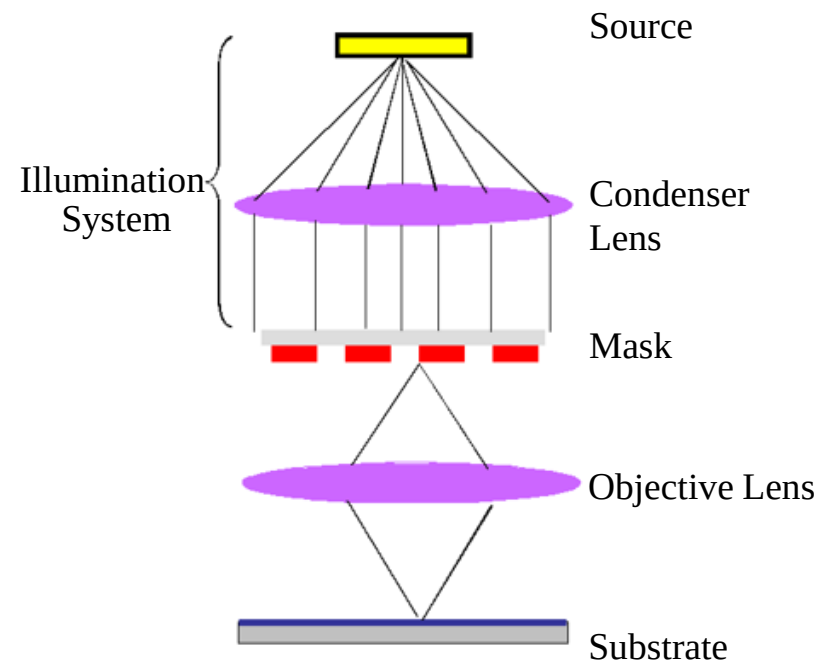
Optical Lithography: Proximity & Projection

Proximity



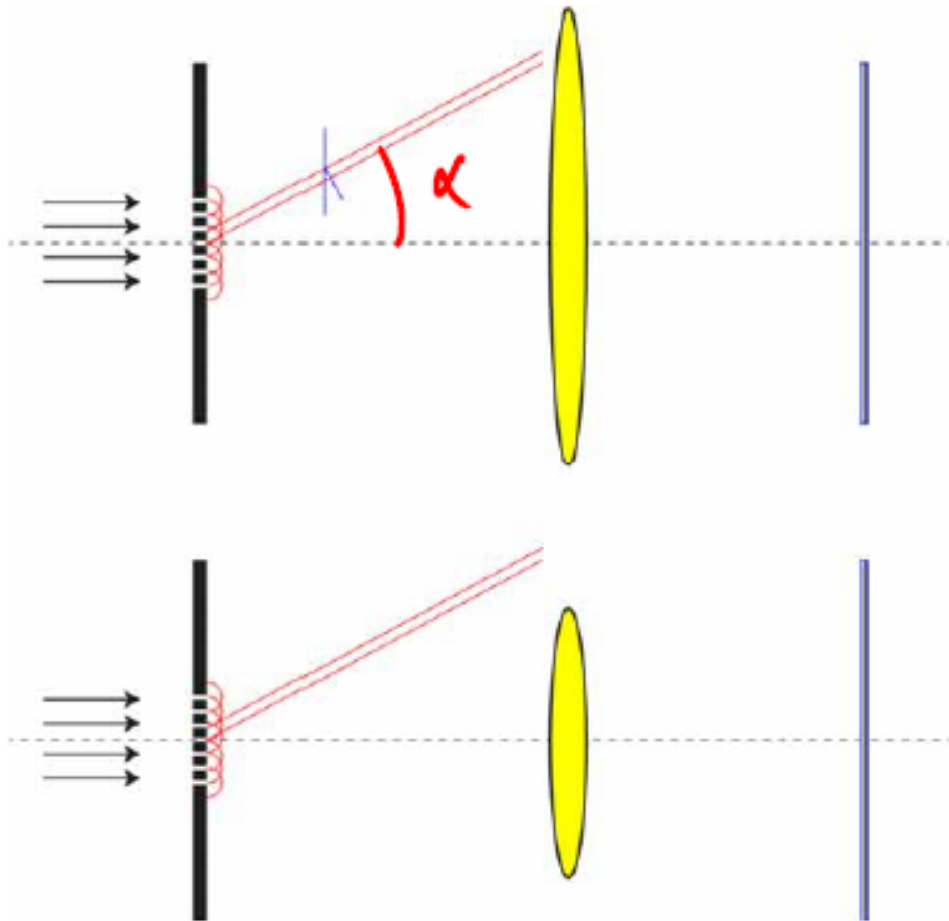
- Simple
- Resolution controlled by λ and z
- Mask issues: 1x, possible damage

Projection



- Complex
- Resolution affected by λ , NA
- Mask 4x, protected

Lithography – Resolution Limits



with 1. order of diffraction:
pattern transfer possible

Resolution:

$$d_{\min} \propto \frac{\lambda}{\sin(\alpha)} = \frac{\lambda}{N.A.}$$

with numerical apertur N.A.



Optical Exposure Systems



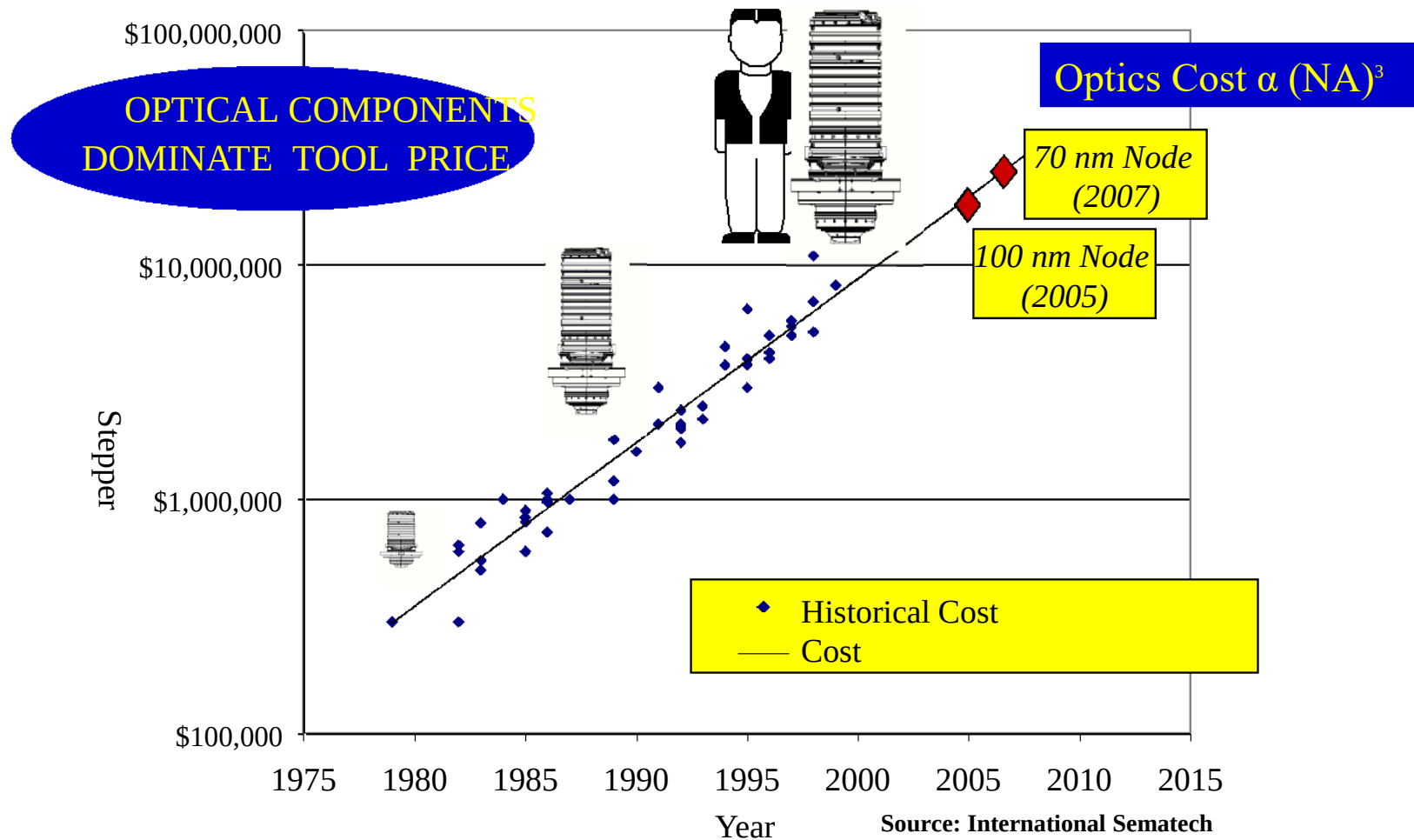
Mask Aligner
(proximity printer)



DUV Stepper
(193 nm)



Cost of Optical Lithography





Lithography Technologies

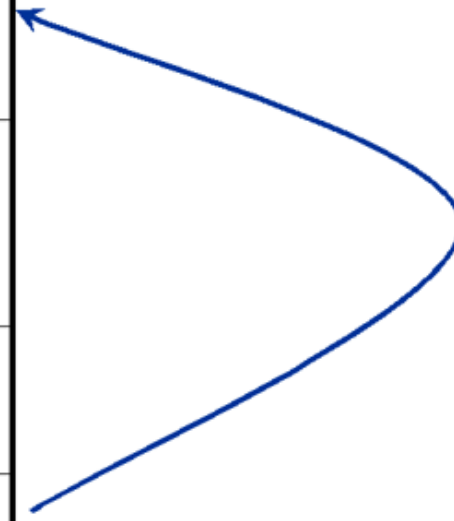
	Resolution (nm)	Placement (nm)	Throughput (features/sec)	Cost (\$)
Optical	~ 50	15	10^{10}	10^7
X-ray	~ 30	15	10^{10}	10^7
EUV	~ 25	10	10^{10}	10^7
E-beam Direct-write	< 10	10	10^4	$10^5 - 10^6$
E-beam Projection	~ 25	20	10^{10}	10^7
Focused Ion Beam	~ 10	10	$10^1 - 10^2$	10^6
Scanning Probe Lithography	~ 30	10	$10^0 - 10^1$	10^5
Micro-contact Printing	~ 50	> 1000	10^{12}	$10^2 - 10^3$
Nanoimprint	< 10	< 100	10^{12}	$10^3 - 10^6$

Lithography Technologies

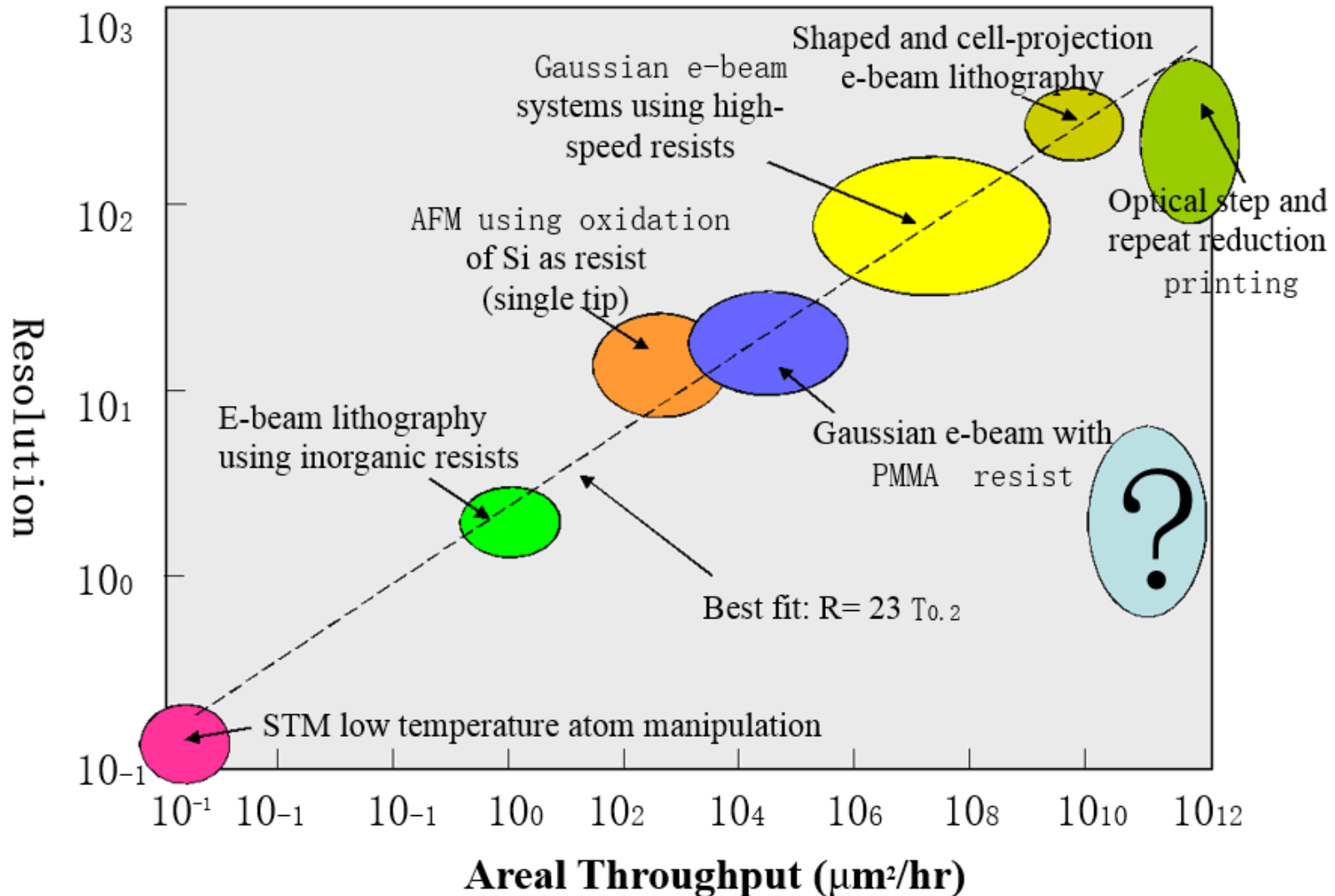
Method	Pros	Cons
Optical proximity	Inexpensive	Poor resolution Contamination
Optical projection	Industry standard	Resolution limited by diffraction Expensive
Electron/ion projection	Resolution	Throughput Shot noise
X-ray proximity	DoF Simple resist	1X mask Source
Imprint	Resolution Throughput	1X mask Overlay
Scanning probe	Resolution	Throughput

Resolution Limiting Factors

Optical	Diffraction
EUV	Secondary e.
E-beam	Probe size Edge slope Scattering
X-ray	Penumbral blur Secondary e.
Ion beam	Probe size Scattering
Nanoimprint	Mask (e-beam)



Resolution vs. Throughput



Contents

- Basic concepts
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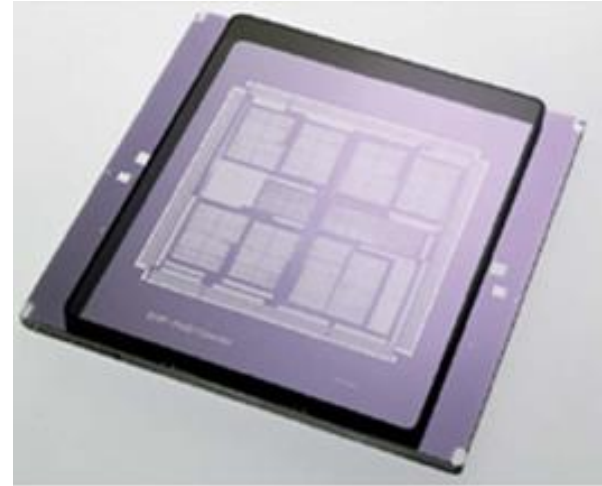




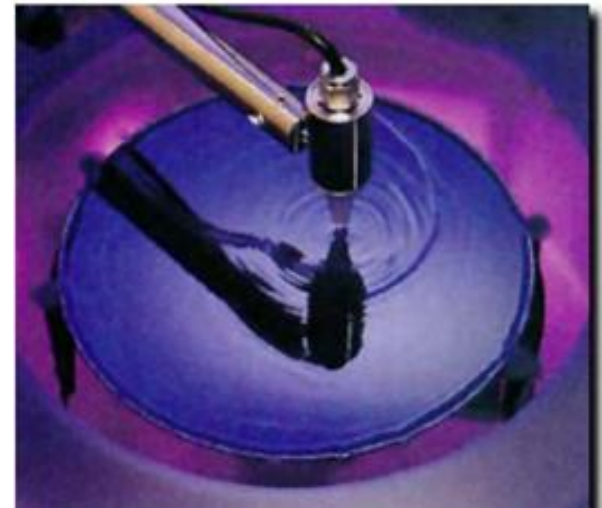
要素之三：Lithographic Materials

- **Mask**

- substrate
- absorber
- scatterer



- **Resist**



Mask Components

- Substrate Requirements

- High transmission
- Low thermal expansion coefficient
- Flatness
- Minimal non-linearities

Common materials:
Quartz, fused silica,
BSC



- Absorber Requirements

- No transmission at λ
- Good adhesion to substrate
- Durability

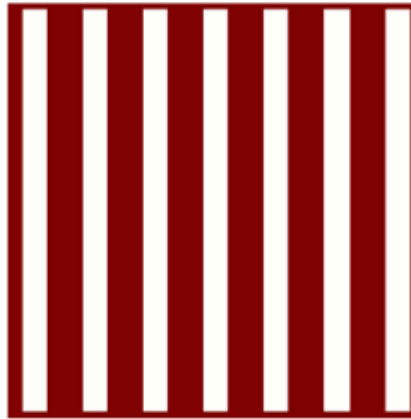
Common materials:
Chrome, emulsion,
 Fe_2O_3

Mask Polarity

Dark field



Grating



Light field



Less adjacent/background exposure

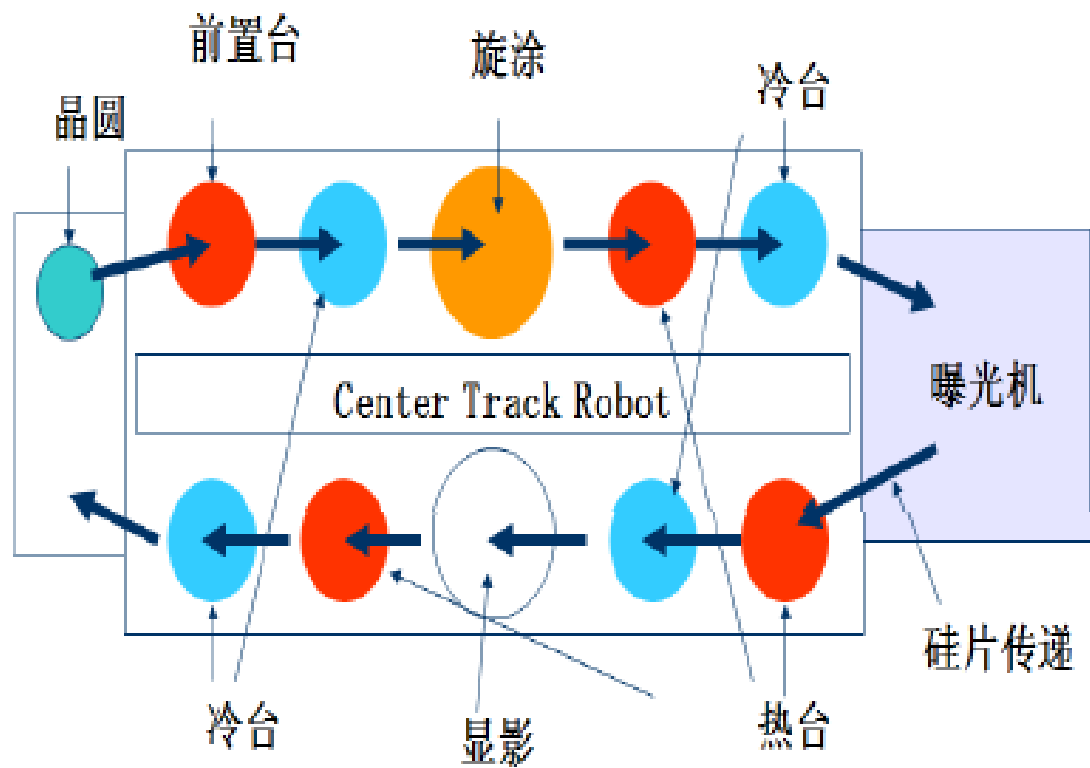
Less impact from defects

Mask patterning???



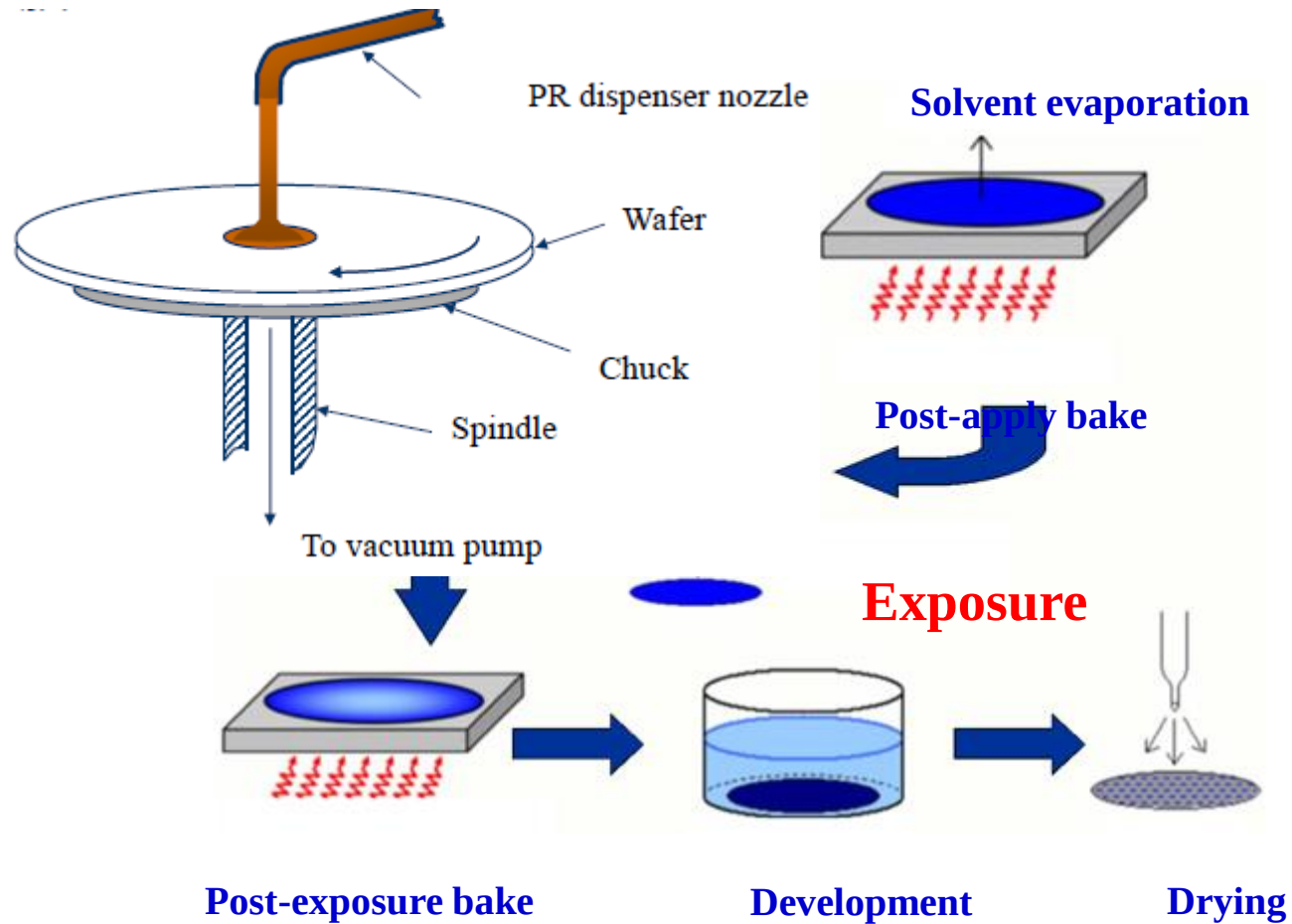
Lithography process

- 1) 气相成底膜处理
- 2) 旋转涂胶
- 3) 软烘
- 4) 对准和曝光
- 5) 曝光后烘焙
- 6) 显影
- 7) 坚膜烘焙
- 7) 显影后检查





Resist Processing



Resist Processing

Resist Thickness:

- Spin speed
- Resist concentration
- Molecular weight (viscosity)

$$T = \frac{KC^B\eta^\gamma}{\omega^\alpha}$$

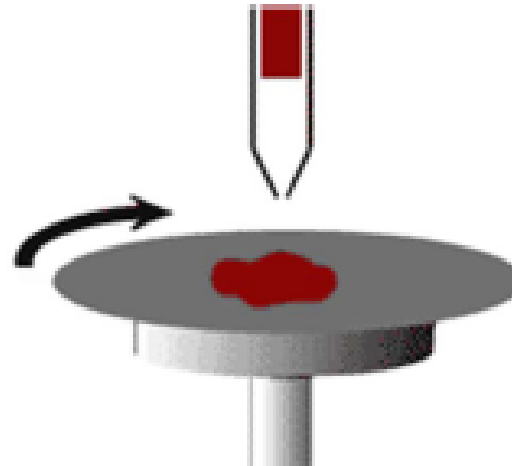
K = calibration constant

C = polymer concentration (g/ml)

η = intrinsic viscosity

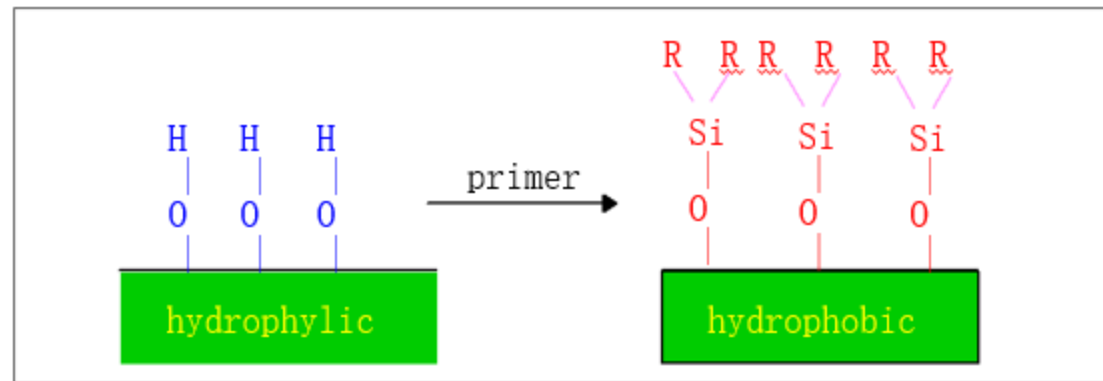
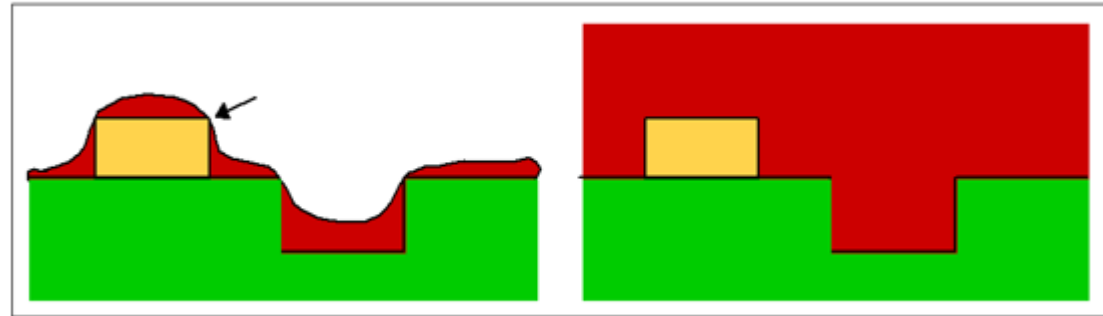
ω = rpm

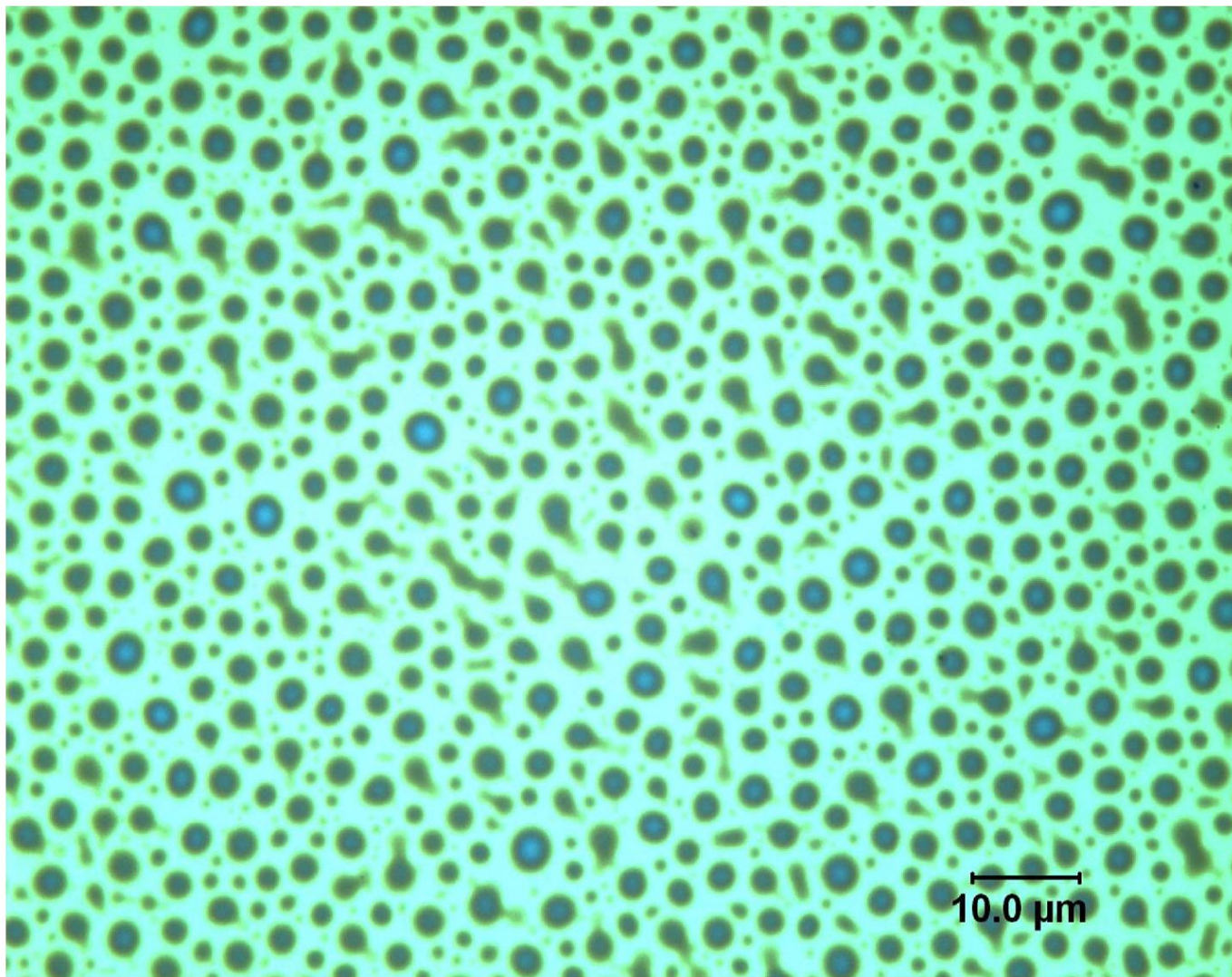
B, γ , α = Empirical factors



Resist Coating

- **Spin process**
 - Viscosity
 - Spin speed
 - Surface
 - Direct ambient
- **Step coverage**
- **Adhesion**





Resolution Limits

Resist & Process

- Intrinsic
 - Polymer size (radius of gyration)
 - Diffusion
- Process
 - Thickness
 - Effective contrast

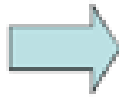
Resist Requirements

- **Contrast**
- **Resolution**
- **Sensitivity**
- **Resistance**
 - to acids
 - to bases
 - to plasmas
 - to ion bombardment
 - to RIE
- **Adhesion to substrates**
- **Homogeneity**
- **Ease of removal Process repeatability**
- **Low defect density**
- **Topography coverage**
- **Temperature stability**
- **Shelf life**
- **Purity from contaminants**

Resist Factors

- **Materials**

- Glass temperature T_g
- Molecular weight
- Substrate atomic number (Z)
- Chemical composition



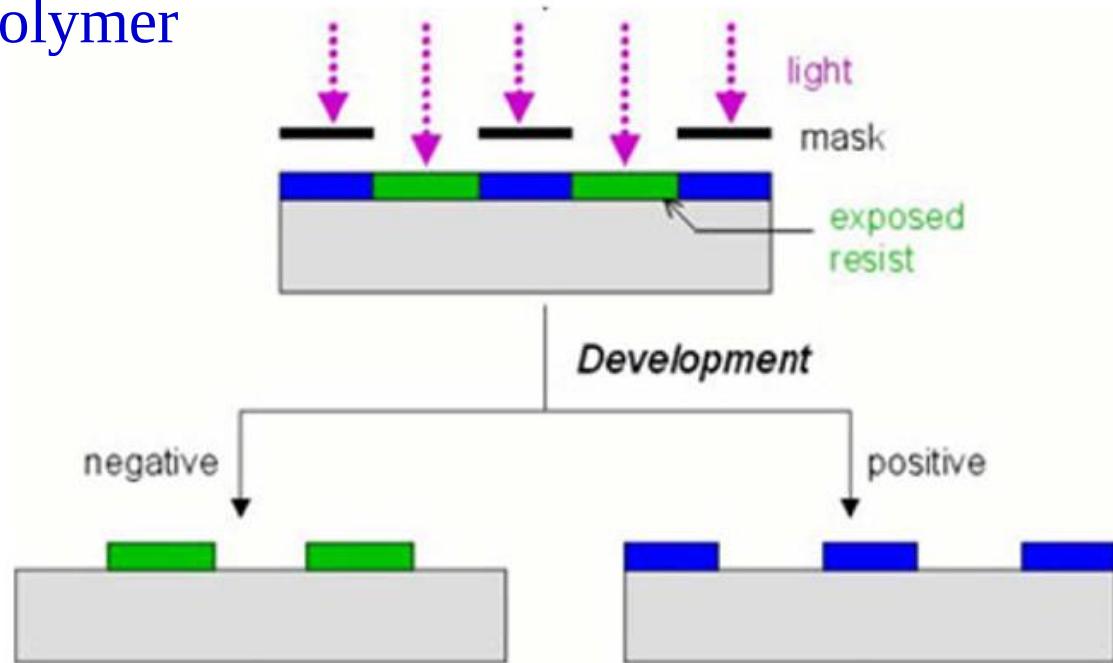
- **Pattern stability**
- **Resolution**
- **Scattering effects**
- **Etch resistance**
- **Adhesion**

- **Process**

- Development (concentration, time, temperature)
- Baking (time, temperature)
- Post-treatment: descum, native oxide strip, etc.

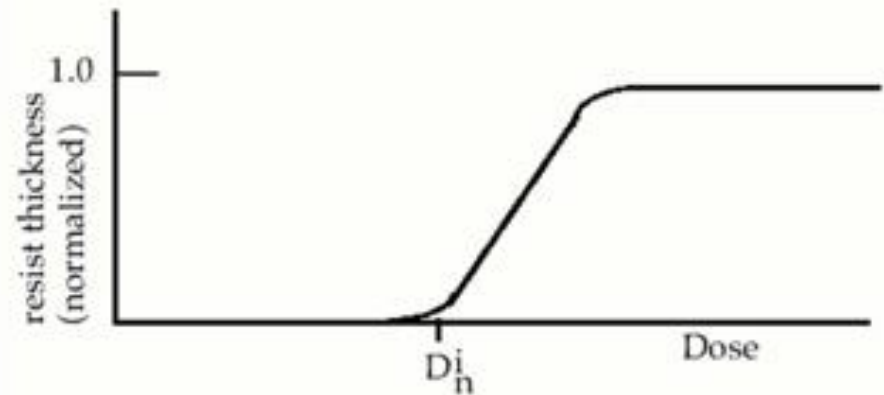
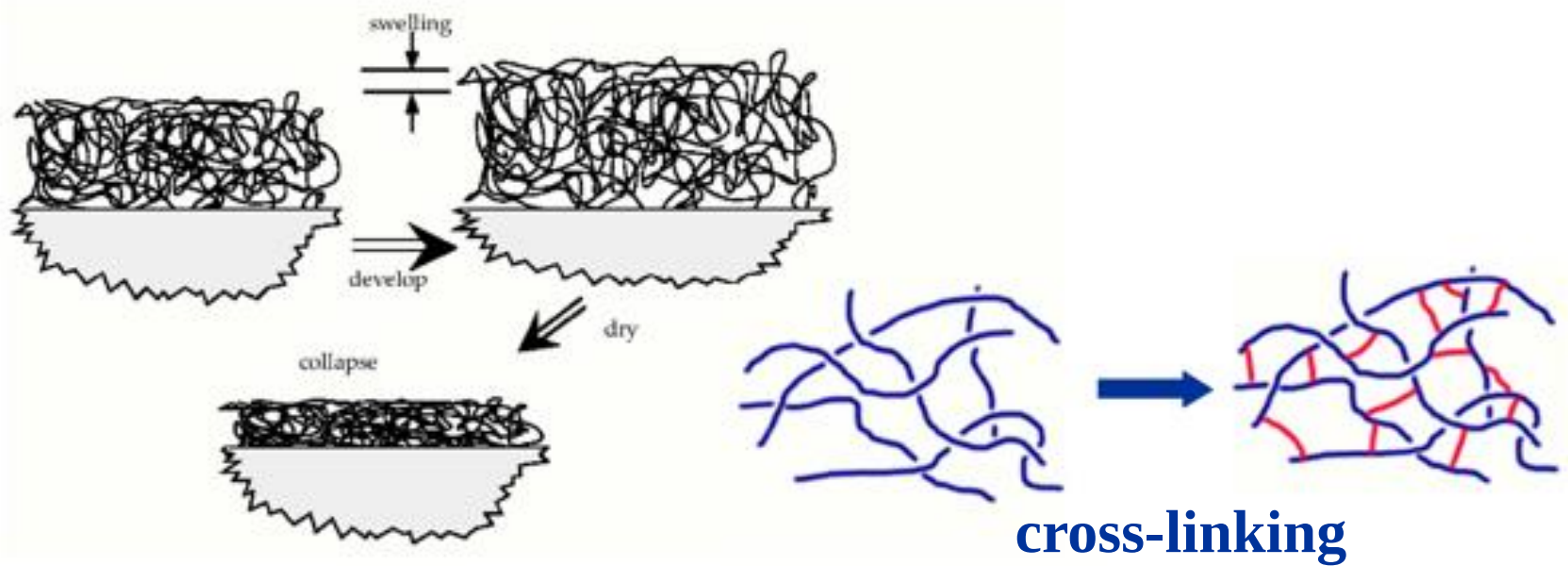
Types of Resists

- Negative cross-linking polymer
- Positive novolak/diazide (线型酚醛清漆)
- Positive chain-scission polymer
- Self-developing
- Chemically amplified
- Sylilated (甲烷硅基化的)



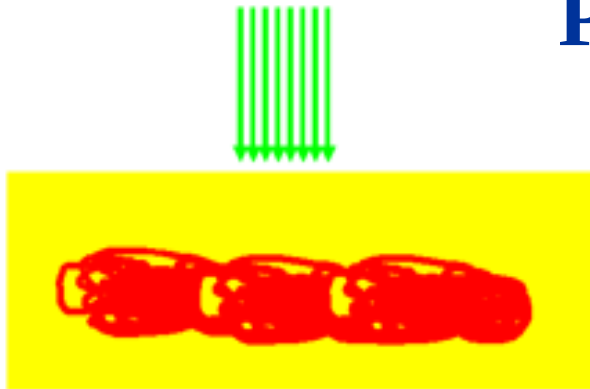


Negative Resist





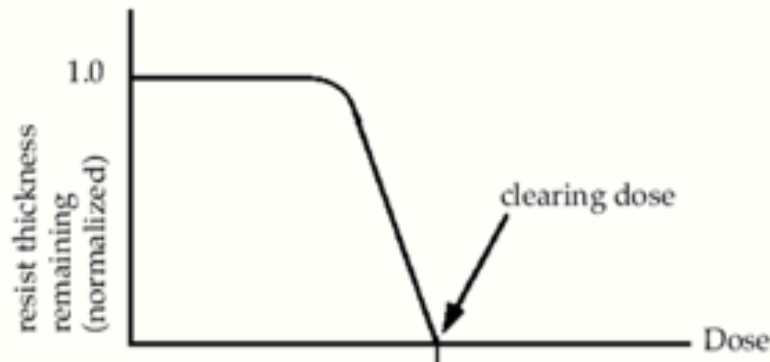
Positive Resist



Chemical modification
during exposure



**Photoactive compound
bleaching or
Chain scission of
matrix polymer**

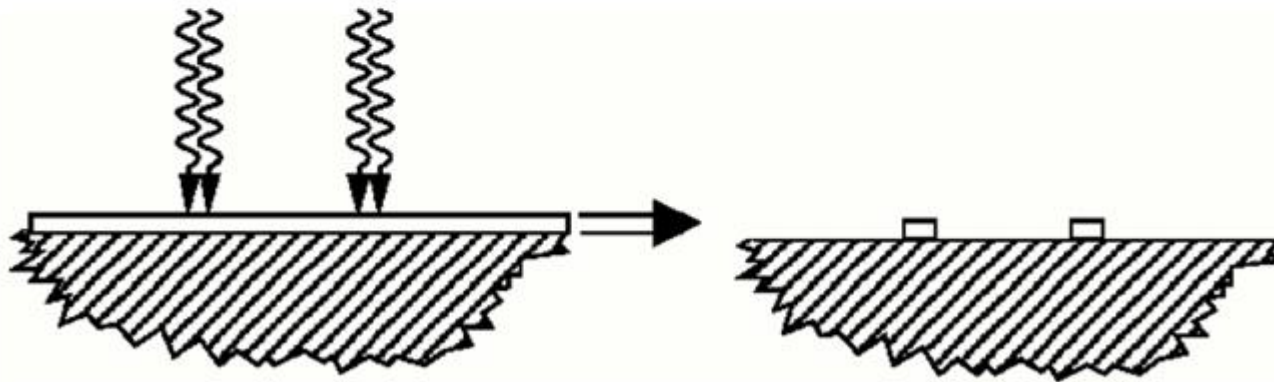
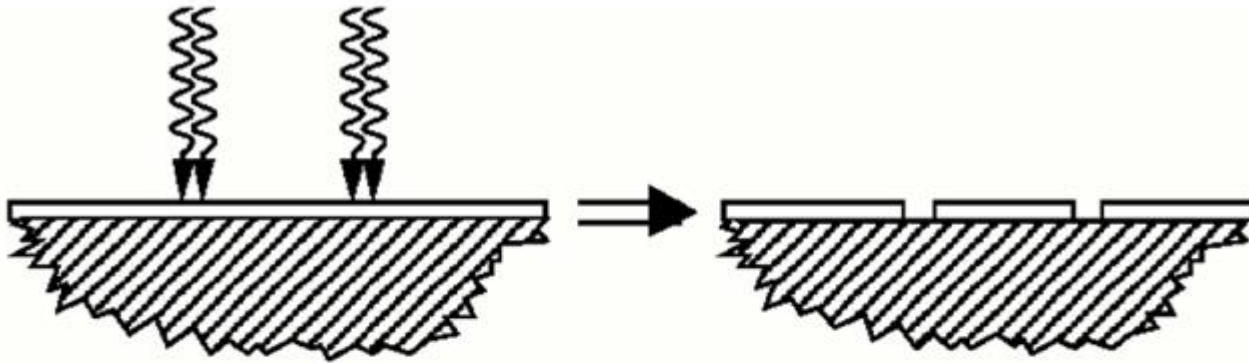


Development





Resists



Resist Components

Matrix Material

- Resin
- Acts as binder
- Adheres to substrate
- Radiation insensitive
- Dissolves rapidly in developer
- Physical or chemical resistant

Sensitizer

- Photoactive compound
- Inhibits dissolution w/o radiation
- Radiation converts it from dissolution inhibitor to dissolution enhancer

Solvent

- Keeps resist in liquid state
- Allows spin coating
- Concentration determines film thickness

Role of Sensitizer (Photoactive Compound)

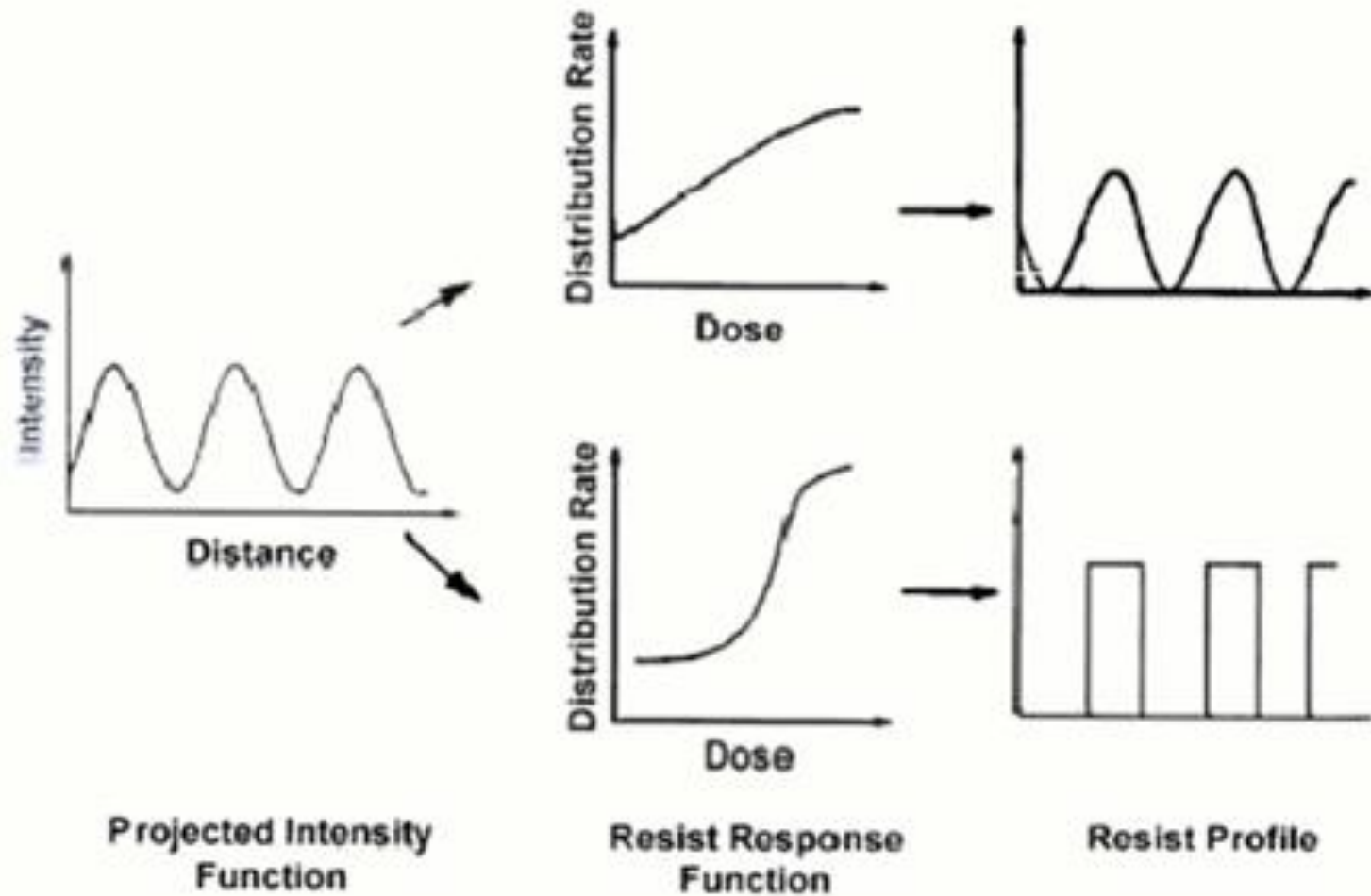
Material	Dissolve Rate in Developer	Function of PAC
Matrix	150 A/s	NA
Matrix + Sensitizer without Radiation	10 – 20 A/s	Dissolution Inhibitor
Matrix + Sensitizer with Radiation	1000 – 2000 A/s	Dissolution Enhancer

solubility $\sim$ 100:1

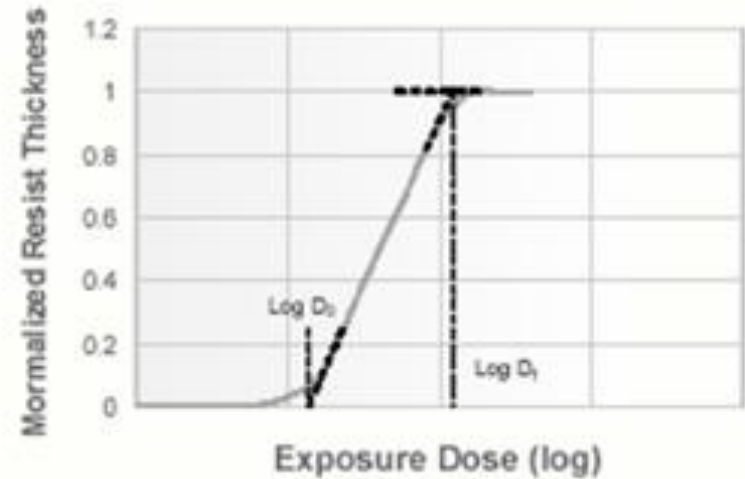
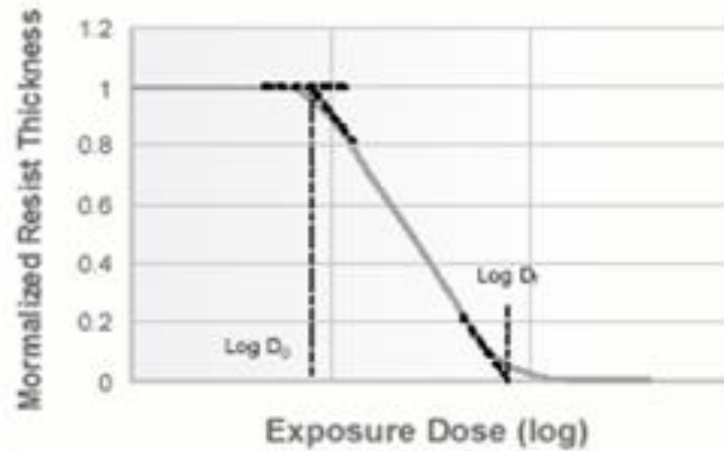


Resist Response

Non-linear dissolution



Resist Response



D_0 = Initial dissolution dose

D_f = 100% dissolution dose

Ideally,
 $D_f \sim D_0 \rightarrow$ High contrast
 D_f small – High sensitivity

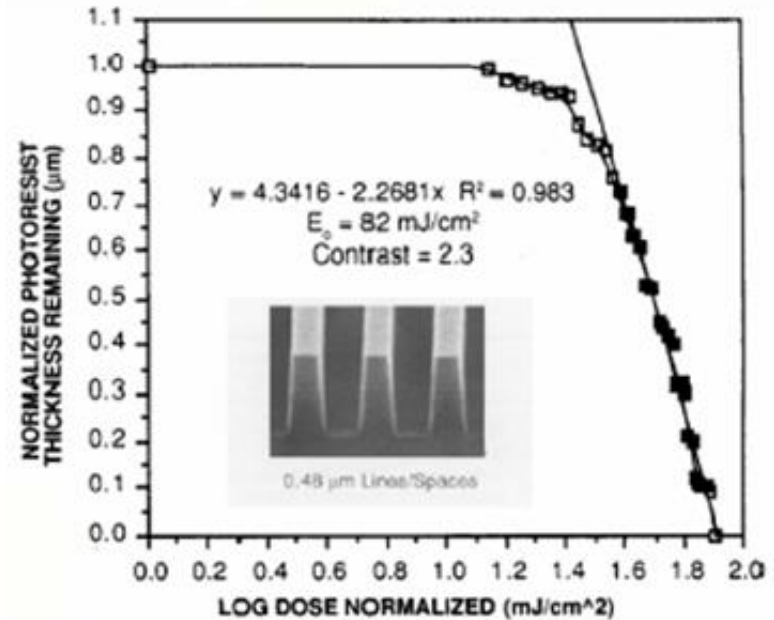
Contrast

$$\lambda = [\log(D_f/D_0)]^{-1}$$

= Ability of resist to distinguish between unexposed and exposed regions

Depends on:

- Resist chemistry
- Radiation
- Processing



Resist Sensitivity

Intrinsic sensitivity: $\Phi = \frac{\text{no. chemical reactions}}{\text{no. absorbed photons/e-s}}$

- Resist's response to radiation
 - Spectral response: Strong absorption $\rightarrow$ high sensitivity
- High sensitivity often correlates with poor resolution

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Key Factors in Optical Lithography



- Resolution (分辨率)
- Resist response (响应)
- Contrast (对比度)
- Placement accuracy (i.e., overlay) (套刻精度)
- Depth of Focus (焦深)

Resolution: Optical

Fraunhofer Diffraction:

(far field projection)

Rayleigh:

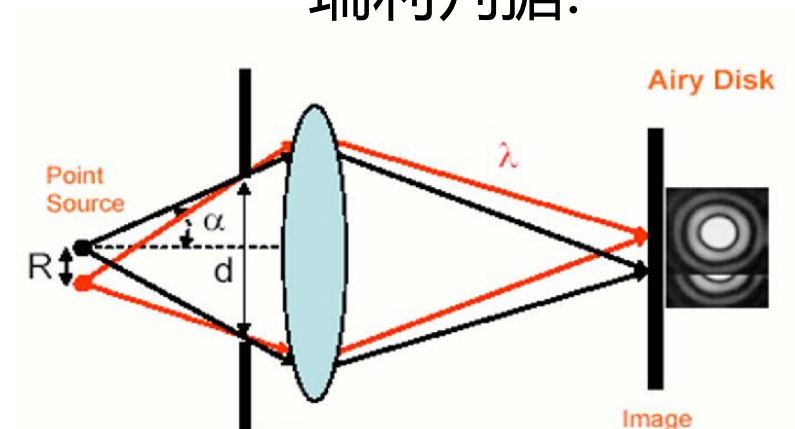
Central maximum of each point source lies at first minimum of Airy disk.

Rayleigh Criterion:

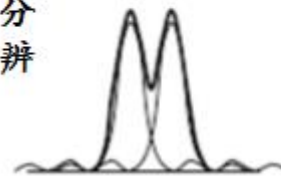
$$R = 0.61\lambda / NA$$

$NA = n\sin(\alpha) = \text{numerical aperture}$

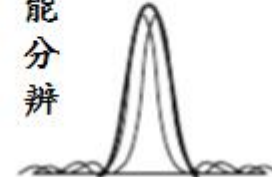
瑞利判据:



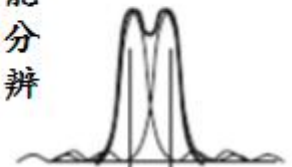
能分辨



不能分辨



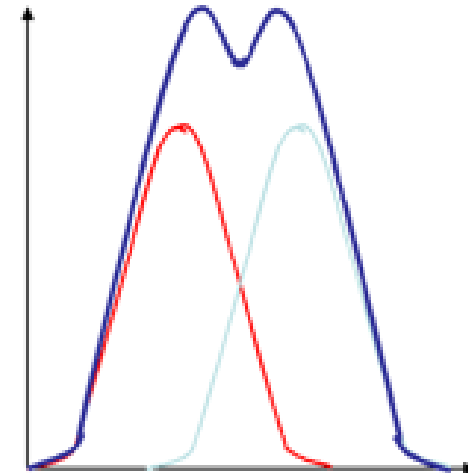
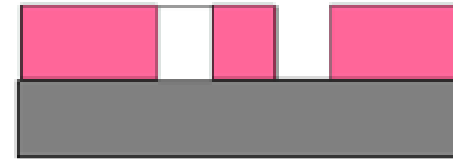
恰能分辨



Contrast in Optical Projection

• Contrast

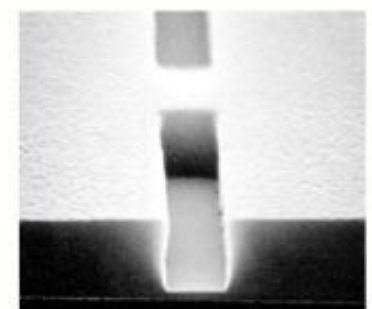
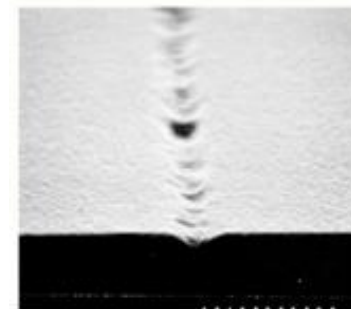
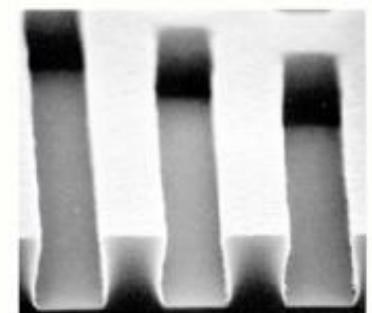
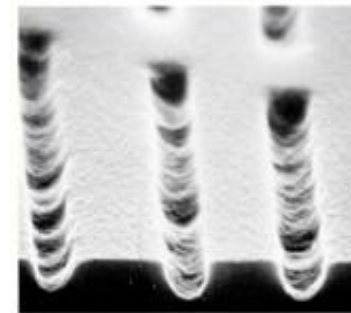
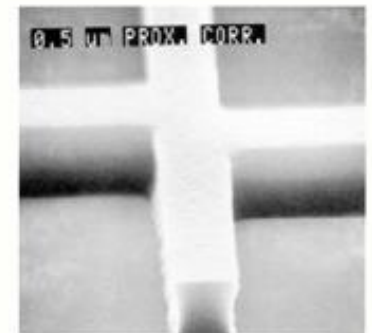
- Depends on required contrast
- What's required?



无衍射效应时, $MTF=1$; 有衍射效应时, $MTF<1$ 。
图形尺寸越小, MTF 越小; 波长越短, 则 MTF 越大

CD(临界尺寸) Control

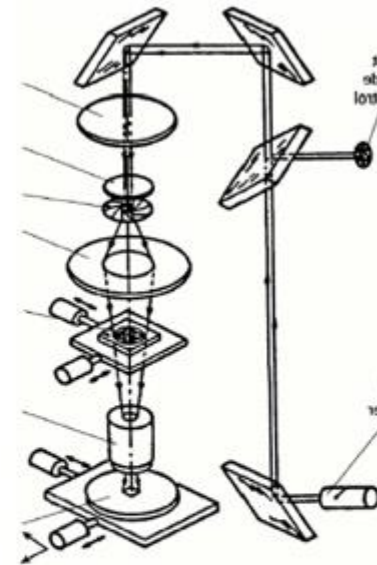
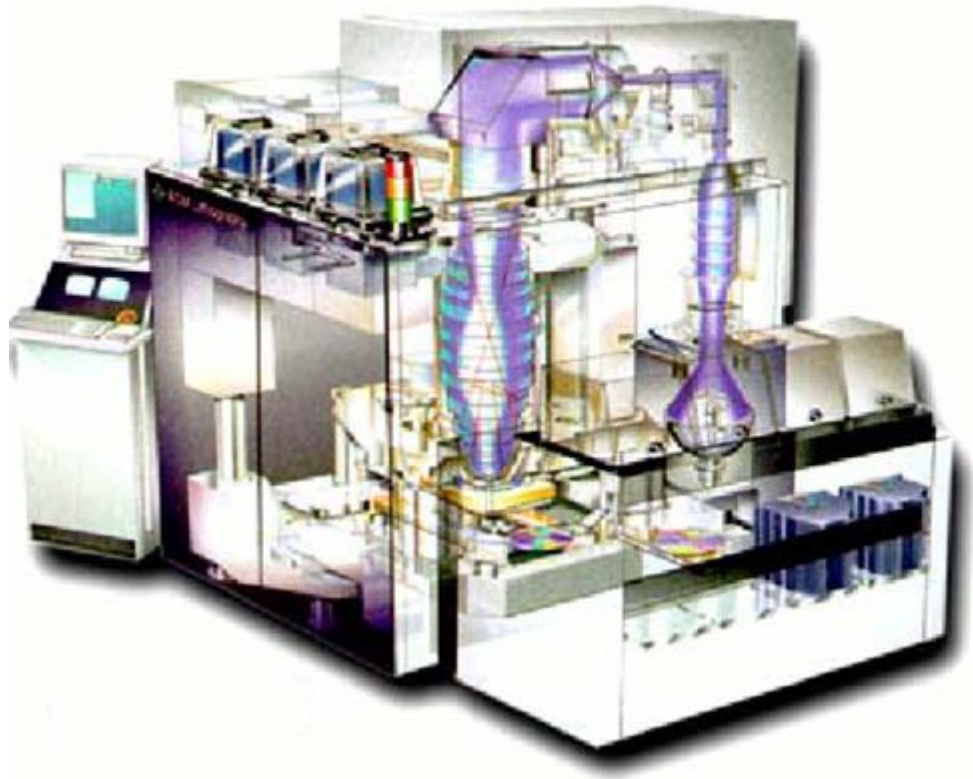
- Exposure uniformity
 - Tool-dependent
 - Illumination
 - Pattern dependent
 - (e.g., proximity effects)
- Resist
 - Thickness
 - Composition
- Developer
- Bake
- Temperature
 - Exposure
 - Development
- Pattern transfer



Factors Affecting Throughput

- Pixel exposure rate
 - flash rate
 - pixels per flash
- Resist sensitivity

Excimer Laser Stepper/Scanner



Light is pulsed

- 20 nsec/pulse

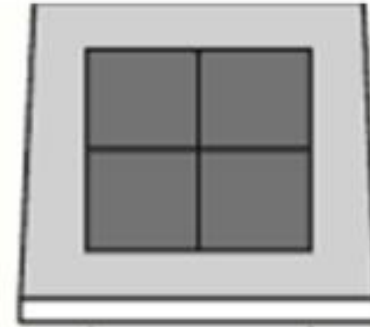
- ~ 50 pulses

- Rep rate ~ a few kHz

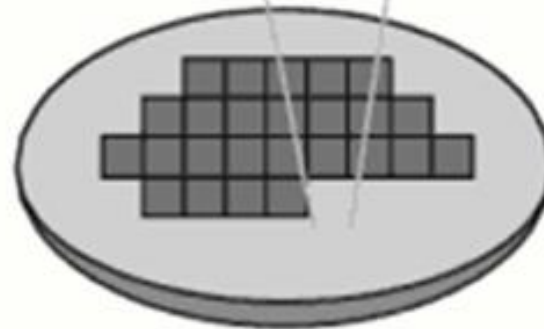


Optical Stepper

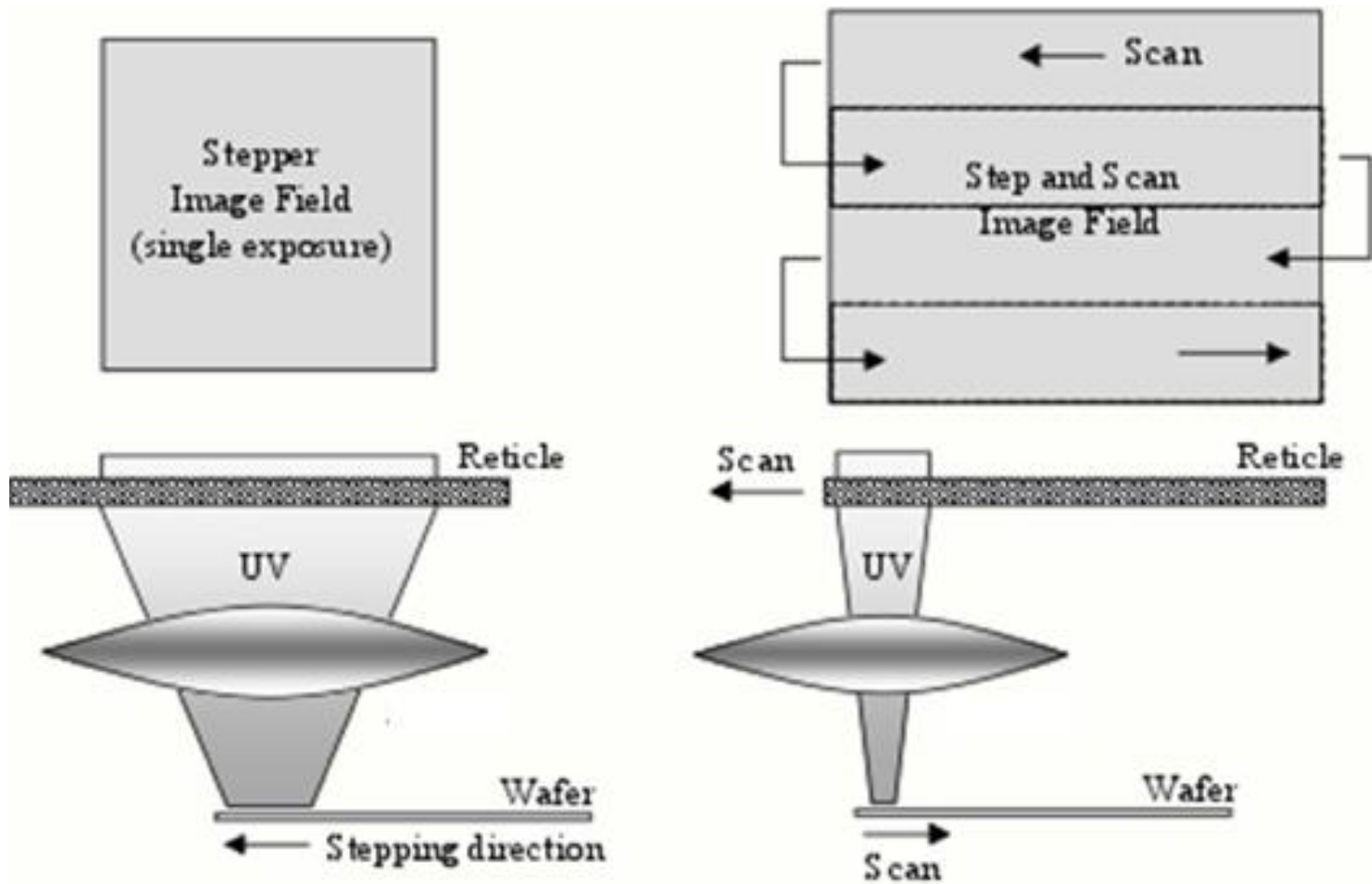
Mask



Wafer



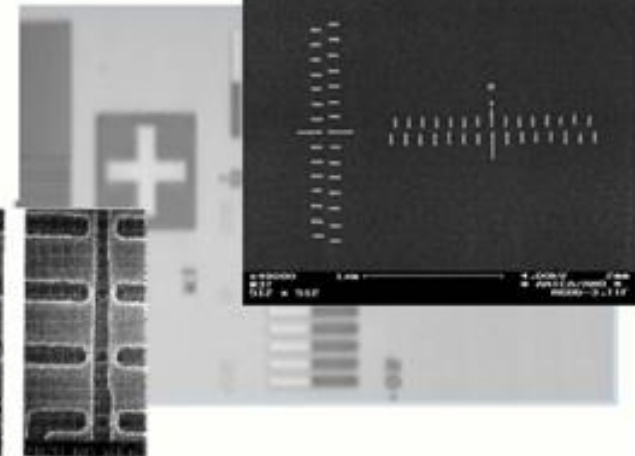
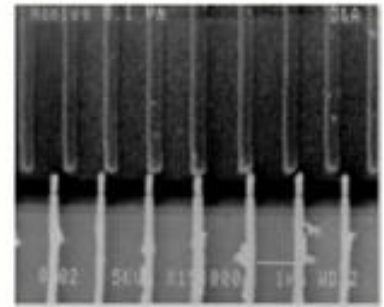
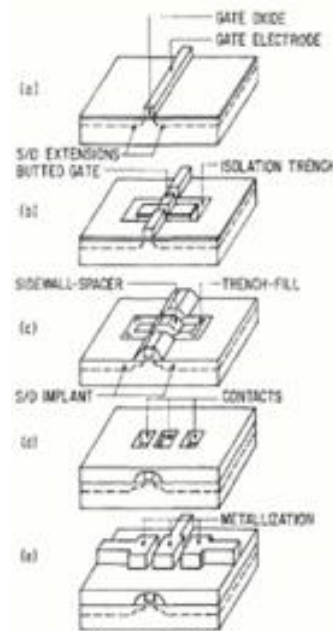
Step and Scan





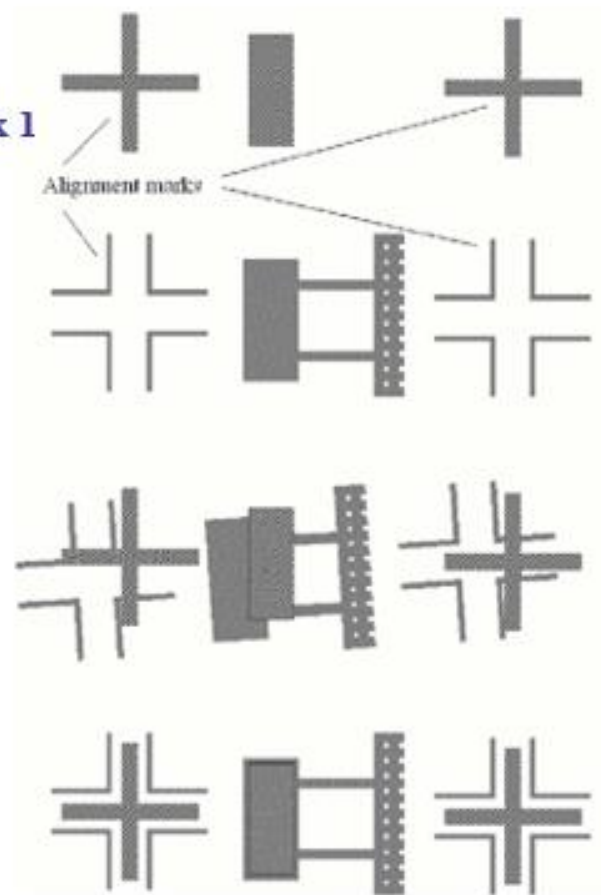
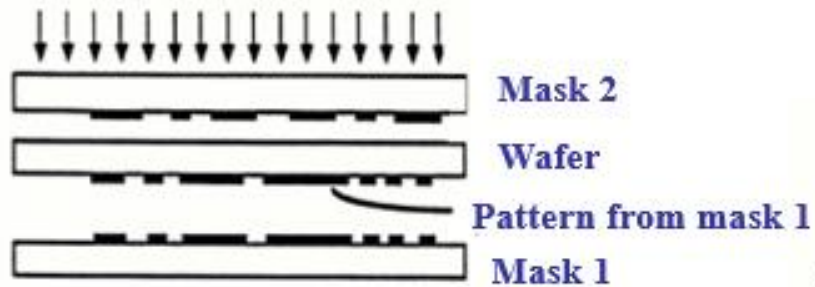
Placement Accuracy

- Layer to layer
 - Parallel writing
 - Serial writing
- Feature to feature
 - Serial writing





Alignment and Overlay



Marks on wafer

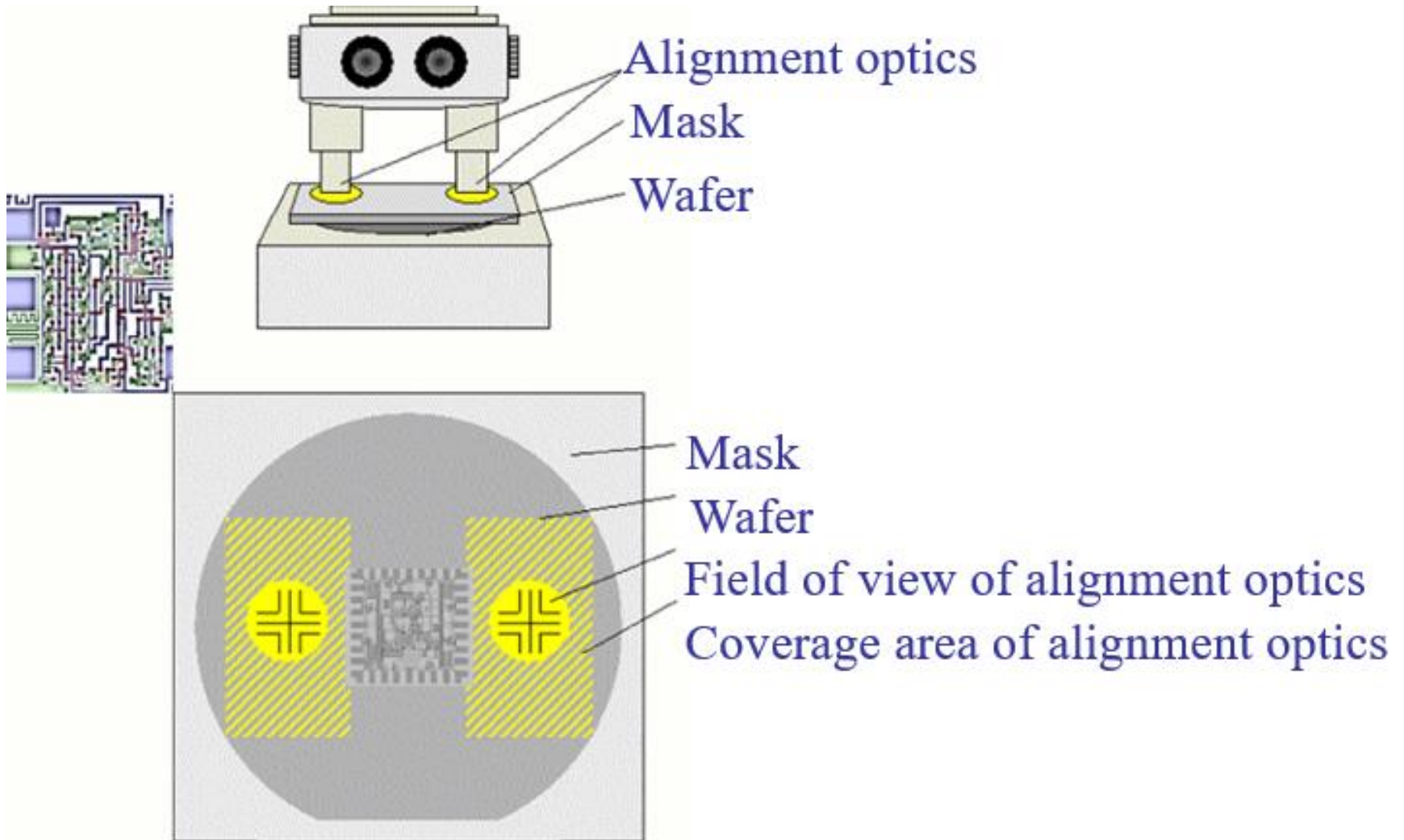
Marks on mask

Mask over wafer

Ready for exposure

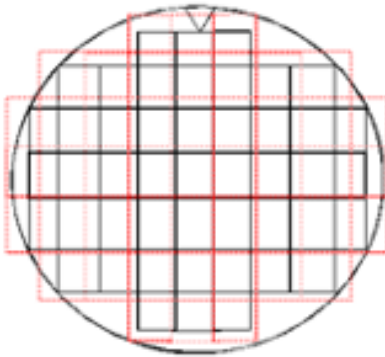


Alignment and Overlay

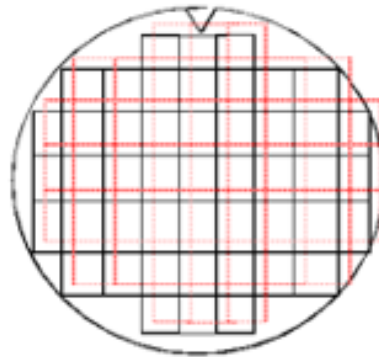




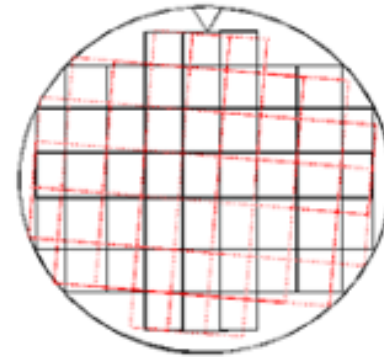
Alignment/Registration Errors



Magnification errors



Translation errors



Rotation errors

F₁₂

Misalignment

- Translation
- Rotation

F₁ **F₂** **F₁** **F₁** **F₁₂** **F₁₂** **F₁₂**
F₂ **F₂** **F₁₂** **F₁₂** **F₁₂**

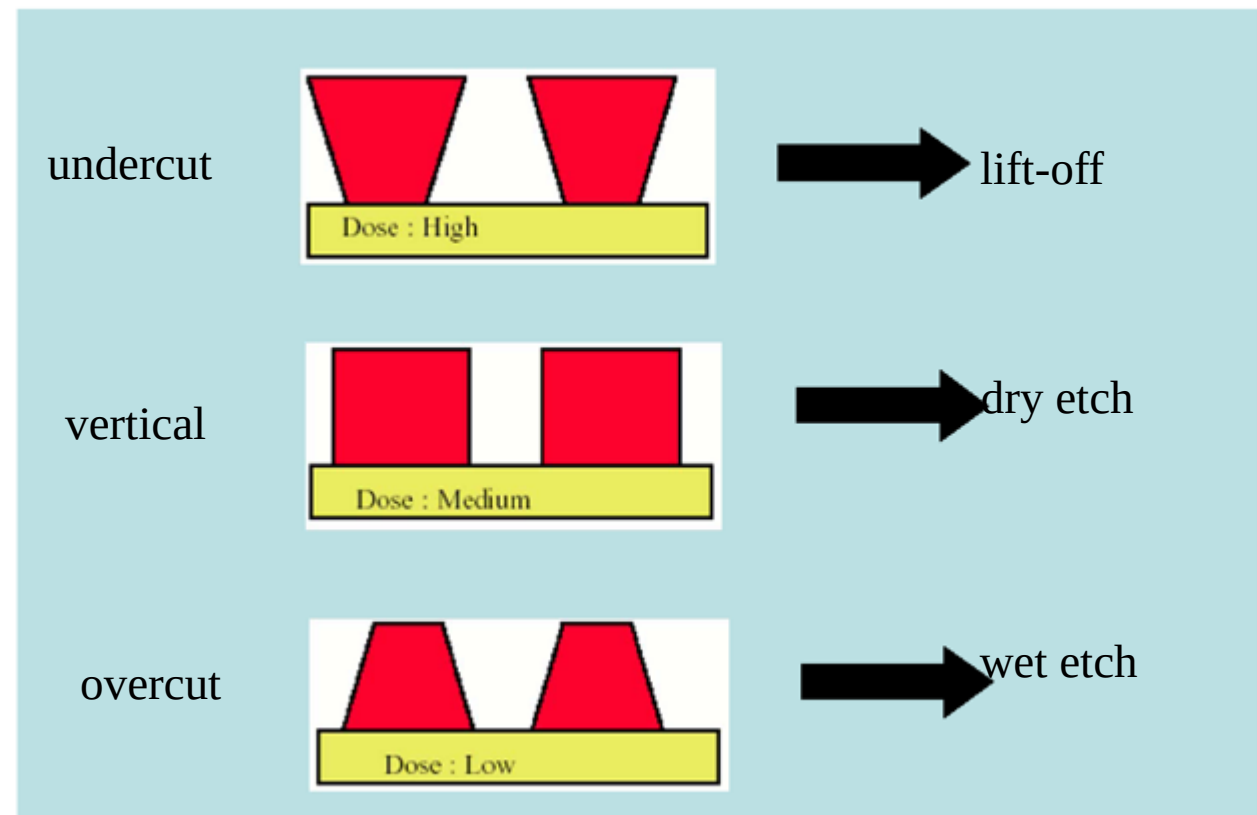
$$\text{Runout} = r [\Delta T_m \alpha_m - \Delta T_w \cdot \alpha_w]$$

r = wafer radius

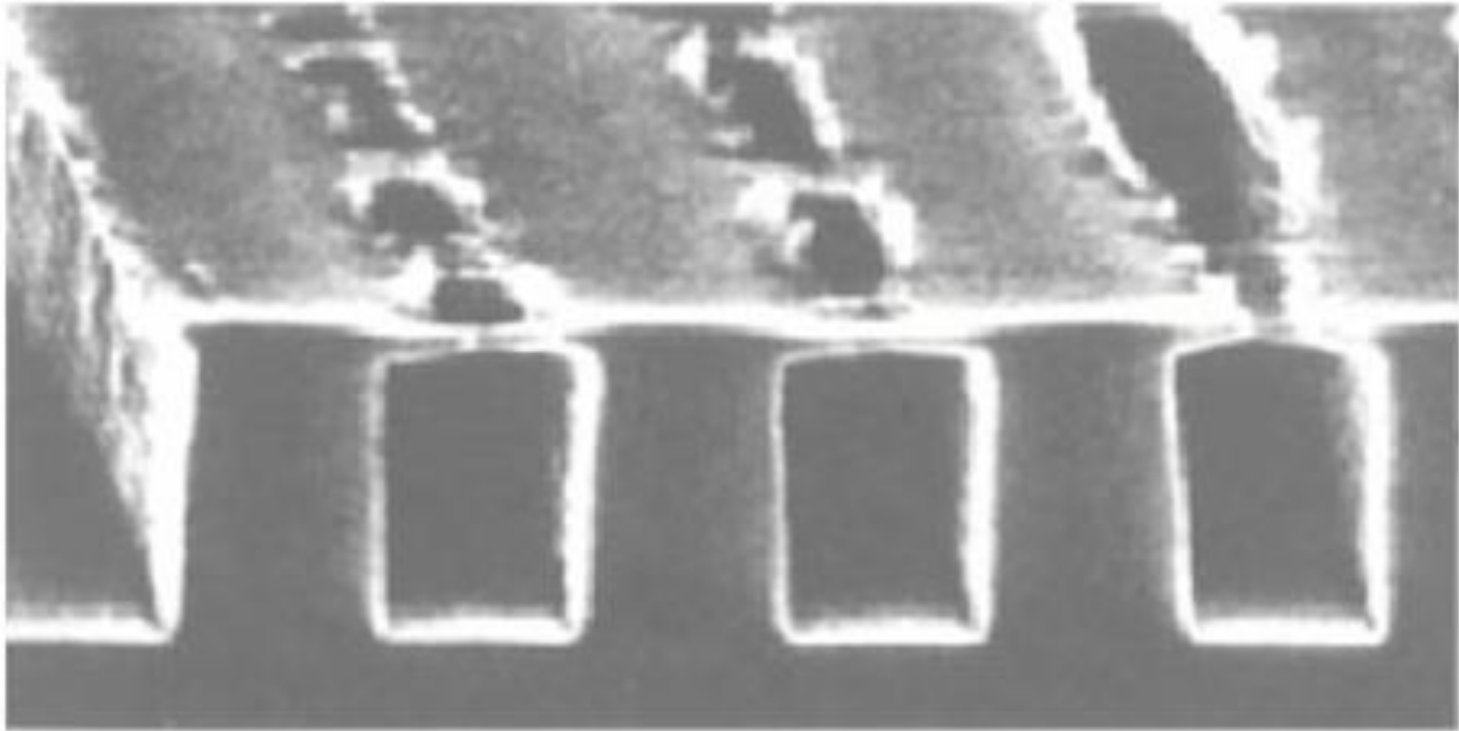
α = coefficient of thermal expansion

Resist Profiles

- Depends on exposure and development
- Can facilitate (or hinder) subsequent processing



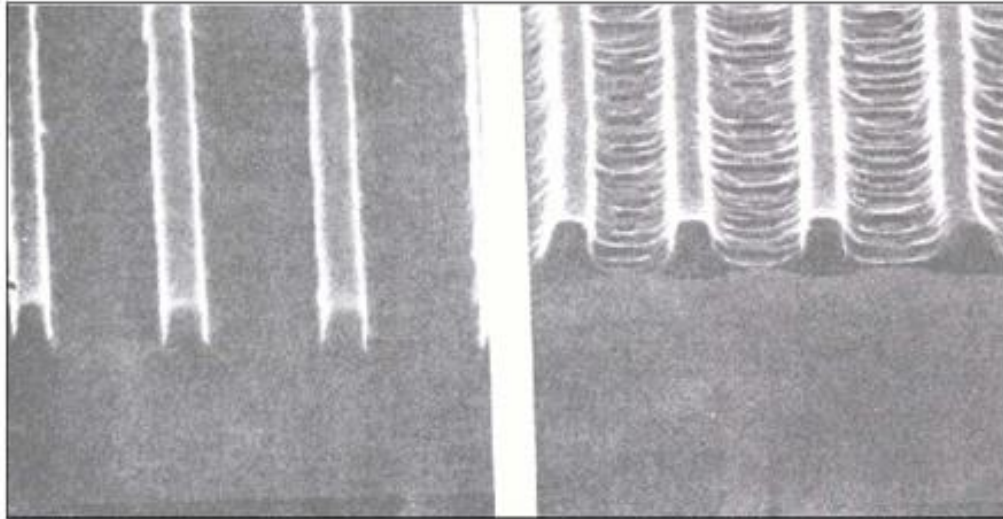
Resist Contamination



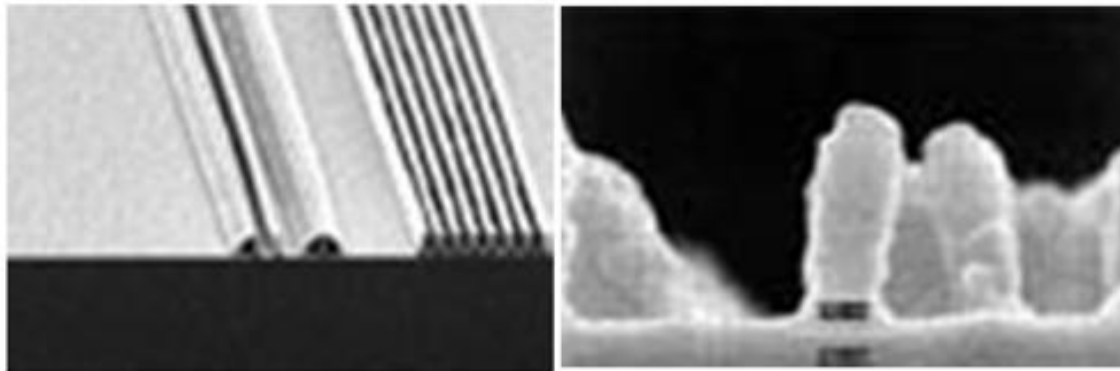
Top surface of DUV resist contaminated by exposure to base



Other Resist Issues



scumming



pattern
collapse



Multilayer Resists

